TSGD: The Sister of SGD

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Abstract

Stochastic gradient descent (SGD) estimates gradients. Transported SGD (TSGD) estimates their changes and uses them to update an initial gradient estimate along the optimization trajectory. Estimating changes can be much easier than estimating gradients: even when individual gradients are highly noisy, shared additive noise cancels exactly from their differences, giving exact transport. With exact initialization as well, TSGD follows gradient descent exactly, regardless of the noise magnitude. This known recursion appears in several variance-reduced methods; here we study it directly as a fundamental estimation mechanism. Its accuracy depends on two error sources: initialization error, which persists, and transport errors, which accumulate. Centered similarity controls the latter through noise variation rather than pointwise noise magnitude. We give a clean, unified convergence analysis that separates these contributions, and derive exact quadratic dynamics showing how the estimator can vanish while the true gradient does not. Simple examples and least-squares experiments identify regimes favoring TSGD or SGD. These results clarify both the value of pure transport and why modern optimizers combine it with fresh estimation.

Is that a shortcoming we should mention in the abstract?

1 Introduction

Stochastic gradient descent (SGD) estimates gradients afresh. Transported SGD (TSGD) estimates gradient changes and uses them to update an existing estimate. SGD is a cornerstone of stochastic optimization and a workhorse of machine learning: replacing the exact gradient with a stochastic estimate makes each update simple and scalable [3]. TSGD follows an equally natural principle. Starting from an initial gradient estimate, it estimates the change between consecutive iterates and adds that change to its current estimate, transporting gradient information along the optimization trajectory.

Estimating changes can be much easier than estimating absolute quantities. Two stochastic gradients evaluated with the same sample can share a noise component; this shared component cancels when they are subtracted, while noise that differs between the two evaluations may remain. In particular, additive noise that is shared across evaluations cancels exactly from their gradient difference, even when individual gradients are highly noisy. The difference then gives the exact gradient change. With exact initialization as well, TSGD follows gradient descent exactly, regardless of the noise magnitude. This is a pathwise identity, not merely a statement in expectation.

The two methods therefore depend on different aspects of the noise. Classical SGD guarantees depend on its pointwise magnitude, commonly bounded by a variance parameter σ^2 [8]. TSGD guarantees, on the other hand, depend on how the noise varies between the coupled evaluations along the trajectory. Centered δ-similarity quantifies this variation: small δ means that stochastic gradient changes closely match deterministic gradient changes. Thus, transport can remain accurate even when pointwise noise is large. This condition applies to population objectives and not only to finite-sum objectives.

Maybe start that sentence with "We stress that this condition applies ..." or a similar beginning.

Transport nevertheless has two fundamental error sources. Initialization error persists, while errors in subsequent gradient changes accumulate. Accurate differences preserve a good initial

True but the phrasing is slightly off.

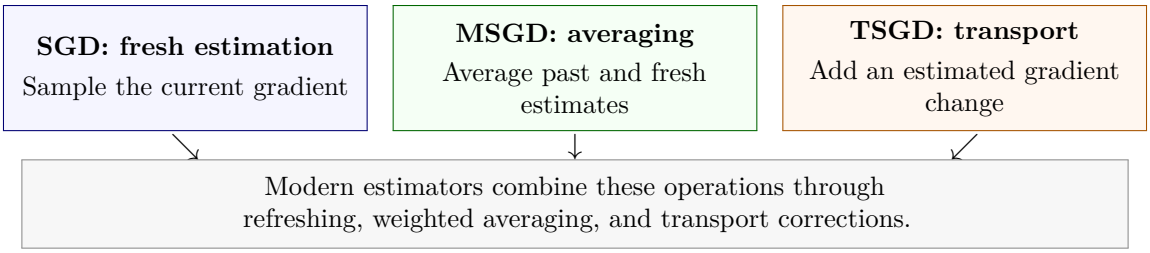


Figure 1: Three fundamental estimation mechanisms. The arrows indicate complementary operations available to modern optimizers, not a requirement that every method use all three.

estimate, but cannot correct an inaccurate one. Even exact initialization does not prevent later transport errors from accumulating. The TSGD guarantee therefore depends on both δ and the accuracy of the initial estimate. At fixed values of these quantities, it is independent of σ, although the noise magnitude can affect the cost of obtaining an accurate initial estimate.

Least-squares regression provides a concrete example. Label noise enters the stochastic gradient $u(u^\top x - b)$, but cancels from the shared-sample difference $uu^\top(x - y)$. It therefore affects fresh gradient estimates and TSGD's initialization, but not its subsequent transport increments. Feature variability can still make those increments noisy (Section 2.2). Do we really need that sentence?

The transport recursion is not new. It appears in established variance-reduced algorithms such as SARAH, SPIDER, and PAGE [6, 15, 16], and independent single-run analyses already exist [17, 18, 21]. It is commonly encountered as part of methods that combine transport with refreshing or mixing. Our purpose is to study the mechanism directly, making its strengths and limitations as explicit as those of SGD.

Momentum SGD (MSGD) provides a third fundamental mechanism: averaging fresh gradients with past estimates. Section 2 introduces all three estimators, and Section 5 describes how modern optimizers use TSGD and combine transport with fresh gradients and momentum averaging.

The paper develops this viewpoint through three contributions:

- **Fundamental viewpoint.** We present TSGD alongside SGD as a basic way of using stochastic-gradient information: estimating changes rather than absolute gradients. Similarity explains when these changes can remain accurate despite noisy individual gradients.
- **Theory and understanding.** We explain similarity, separate initialization and transport errors, and give a finite-horizon guarantee under conditional assumptions on the gradient change increments. Exact quadratic dynamics quantify their interaction and show how the estimator can vanish while the true gradient does not.
- **Comparison and connections.** Simple examples identify regimes favoring TSGD or SGD. Least-squares experiments show label noise cancelling from transport and a transported estimate vanishing at a nonstationary point. We compare convergence guarantees and oracle costs, and discuss why modern optimizers typically combine transport with fresh estimation and momentum averaging.

Third contribution seems very weak.

The examples precede the convergence analysis so that the role of each assumption is visible before the theorem. Detailed proofs and additional calculations are collected in the appendices.

collected is a weird word here.

2 Three Fundamental Estimators and Similarity

2.1 SGD, MSGD, and TSGD

Why do we even care about that question?

How should we construct an estimate of $\nabla f(x_{k+1})$? The identity

$$\nabla f(x_{k+1}) = \nabla f(x_k) + [\nabla f(x_{k+1}) - \nabla f(x_k)]$$

use "by" instead of a comma imo.

suggests two approaches. SGD estimates the left-hand side directly, drawing a fresh stochastic gradient at the new point. TSGD estimates the right-hand side: its current estimate g_k already approximates the first term, so it needs new information only for the change. Using a stochastic gradient oracle h with the same fresh sample at both endpoints gives

$$g_{k+1} = \underbrace{g_k}_{\approx \nabla f(x_k)} + \underbrace{h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1})}_{\approx \nabla f(x_{k+1}) - \nabla f(x_k)}.$$

Thus TSGD updates the gradient information it already has. The sample ξ_{k+1} is shared within the difference and drawn afresh at each iteration. For a finite sum, this means evaluating the same randomly selected component at both points. TSGD retains the information in g_k, including its estimation error; the effect of these inherited errors is examined in Section 2.3.

We formalize the setup as follows.

Assumption A1 (Smooth, lower-bounded objective)**.** The function $f : \mathbb{R}^d \to \mathbb{R}$ has an L-Lipschitz gradient, with $L > 0$, and $f_{\inf} := \inf_x f(x) > -\infty$.

Mention the key inequality for L-Lipschitz gradients.

Assumption A2 (Unbiased stochastic gradients)**.** The oracle satisfies $\mathbb{E}_\xi[h(x, \xi)] = \nabla f(x)$ for every x. Each new sample is independent of the preceding history.

All three methods use positive deterministic stepsizes η_k and the same iterate update, starting from a deterministic x_0:

$$x_{k+1} = x_k - \eta_k g_k. \tag{2.1}$$

Their gradient estimates differ:

$$\text{SGD}: \quad g_{k+1} = h(x_{k+1}, \xi_{k+1}), \tag{2.2}$$

$$\text{MSGD}: \quad g_{k+1} = (1 - \beta_k) g_k + \beta_k h(x_{k+1}, \xi_{k+1}), \quad 0 < \beta_k \le 1, \tag{2.3}$$

$$\text{TSGD}: \quad g_{k+1} = g_k + D_{k+1}. \tag{2.4}$$

In the standard pointwise-oracle construction,

$$D_{k+1} = h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1}). \tag{2.5}$$

start this paragrph by describing the SGD g_k, not MSGD's.

MSGD averages fresh and past estimates to reduce noise, but can lag behind a changing gradient. TSGD accounts for the gradient change through transport. Unless specified otherwise, SGD and MSGD initialize with $g_0 = h(x_0, \xi_0)$. TSGD permits any g_0 with finite mean-square error, including a biased estimate.

2.2 Understanding Similarity

Write $r(x, \xi) = h(x, \xi) - \nabla f(x)$ for the gradient noise.

start off by saying that SGD is typically analyzed under the BV assumption.

Assumption A3 (Bounded variance (BV))**.** For some $\sigma \ge 0$,

$$\mathbb{E}_\xi[\|r(x, \xi)\|^2] \le \sigma^2 \qquad \text{for all } x. \tag{2.6}$$

Transport instead uses a condition on gradient differences.

Assumption A4 (Centered δ-similarity)**.** For some $\delta \geq 0$,

$$\mathbb{E}_\xi[\|r(x,\xi) - r(y,\xi)\|^2] \leq \delta^2\|x-y\|^2 \qquad \text{for all } x, y. \tag{2.7}$$

Under uniform finite-sum sampling, this condition coincides with the Hessian-variance condition of Szlendak et al. [22] and the second-order similarity assumption of Khaled and Jin [13]. More recently, Shulgin et al. [20] generalized it through γ-similarity, which recovers this condition at $\gamma = 1$. We use the name *centered δ-similarity* to emphasize that it measures the variation of gradient noise rather than its magnitude.

Here σ bounds the magnitude of individual gradient errors, whereas δ measures their variation under shared sampling. A small δ means that stochastic gradient differences closely match deterministic gradient differences; it does not require the gradients themselves, or their changes, to be small. Similarity is a condition on the stochastic oracle and applies to both finite-sum and population objectives.

use 1/n instead of n^-1.

For an illustration, let $f = n^{-1}\sum_i f_i$ with $f_i(x) = \frac{1}{2}x^\top A_i x + b_i^\top x + c_i$ and symmetric A_i. Uniformly sampling i, and writing $\bar{A} = n^{-1}\sum_i A_i$, gives

$$[\nabla f_i(x) - \nabla f_i(y)] - [\nabla f(x) - \nabla f(y)] = (A_i - \bar{A})(x-y). \tag{2.8}$$

The linear terms $b_i^\top x$ cancel from the gradient differences. Components can therefore have very different gradients but nearly identical gradient changes. Identical Hessians give exact transport even with arbitrarily large additive gradient noise. Hessian heterogeneity controls δ, whereas L concerns the average Hessian; neither must be smaller than the other (Appendix C).

Scalar least squares separates the two quantities. Let $f(x) = \frac{1}{2}\mathbb{E}[(ux-b)^2]$, with feature u satisfying $0 < \mathbb{E}[u^2]$ and $\mathbb{E}[u^4] < \infty$, and label $b = ux^\star + \varepsilon$. The label noise ε is independent of u, with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = s^2 < \infty$. The stochastic gradient is $h(x;u,\varepsilon) = u^2(x-x^\star) - u\varepsilon$, so

First we have the finite-sum example.
Second we have the least squares example.
Maybe its too much. Maye not.
What do you think sherlock ?

$$\begin{aligned} h(x;u,\varepsilon) - h(y;u,\varepsilon) &= u^2(x-y), \\ \mathbb{E}[|h(x;u,\varepsilon) - \nabla f(x)|^2] &= \delta^2(x-x^\star)^2 + Ls^2, \end{aligned} \tag{2.9}$$

where $\delta^2 = \mathrm{Var}(u^2)$ and $L = \mathbb{E}[u^2]$. The two terms play different roles. Label noise contributes to the noise magnitude but cancels from differences, while feature variability sets δ. Increasing s therefore makes every fresh gradient noisier without affecting transport. When $\delta > 0$, the pointwise noise variance is unbounded in x, so similarity holds but global BV fails. Label noise still affects TSGD through its initial estimate, but not through later transport increments. Sections 3.1 and 3.3 illustrate the two extremes of this model: a constant feature ($\delta = 0$) with label noise favors transport with accurate initialization, whereas noiseless labels ($s = 0$) with varying features make fresh gradients exact at the minimizer.

SGD can also be analyzed under the ABC condition, which generalizes BV [14]. Both constrain the stochastic gradient at each point separately; centered similarity instead constrains noise differences at two points under shared sampling. Appendix C.1 compares these conditions.

2.3 Two Fundamental Sources of TSGD Error

Start off by saying that we earlier mentioned TSGD has 2 sources of errors

Define $e_k = g_k - \nabla f(x_k)$, $E_k = \mathbb{E}[\|e_k\|^2]$, and $d_{k+1} = D_{k+1} - [\nabla f(x_{k+1}) - \nabla f(x_k)]$. The transport update gives

$$\boxed{e_{k+1} = e_k + d_{k+1}.} \tag{2.10}$$

Let $\mathcal{F}_k$ denote the history before drawing D_{k+1}, including x_{k+1} and g_k. Conditionally unbiased increments give

$$\mathbb{E}[g_{k+1} \mid \mathcal{F}_k] = \nabla f(x_{k+1}) + e_k.$$

Unlike SGD, TSGD is therefore generally conditionally biased: the new increment is unbiased, but the estimator inherits its previous error. The increment error d_{k+1} has zero conditional mean and is orthogonal in expectation to previous errors. Consequently,

$$E_k = E_0 + \sum_{t=0}^{k-1} \mathbb{E}[\|d_{t+1}\|^2]. \tag{2.11}$$

Even if $\mathbb{E}[e_0] = 0$, so that $\mathbb{E}[e_k] = 0$, the mean-square error E_k cannot decrease. Initialization error persists, while transport errors accumulate. Similarity bounds each contribution by the squared displacement, yielding

$$E_k \le E_0 + \delta^2 \sum_{t=0}^{k-1} \eta_t^2 \mathbb{E}[\|g_t\|^2]. \tag{2.12}$$

This explains both the opportunity and the difficulty: accurate changes preserve useful information, but even small errors can accumulate. Section 4 states the conditional assumptions directly on D_{k+1}, allowing transport beyond pointwise gradient oracles.

3 When Does TSGD Shine?

The distinction between noise magnitude and noise variation is easiest to see on simple objectives. We compare the expected squared norm of the true gradient, $G_k := \mathbb{E}[\|\nabla f(x_k)\|^2]$, keeping track of the initial estimate as a separate resource.

keeping track... seems weird phrasing. Also, the paragraph should somehow mention that we'll be giving examples/understanding showing when TSGD is useful.

Alongside exact calculations, we use a vector least-squares model $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$ as a running experiment. Its population Hessian is held fixed while feature variability sets δ and label noise varies independently (Section 2.2).

3.1 Accurate Differences Despite Noisy Gradients

Consider $f(x) = x^2/2$ and an unbiased oracle $h(x, \xi) = x + \xi$, where $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \sigma^2$. The individual gradients can be arbitrarily noisy, yet

$$h(x, \xi) - h(y, \xi) = x - y = \nabla f(x) - \nabla f(y).$$

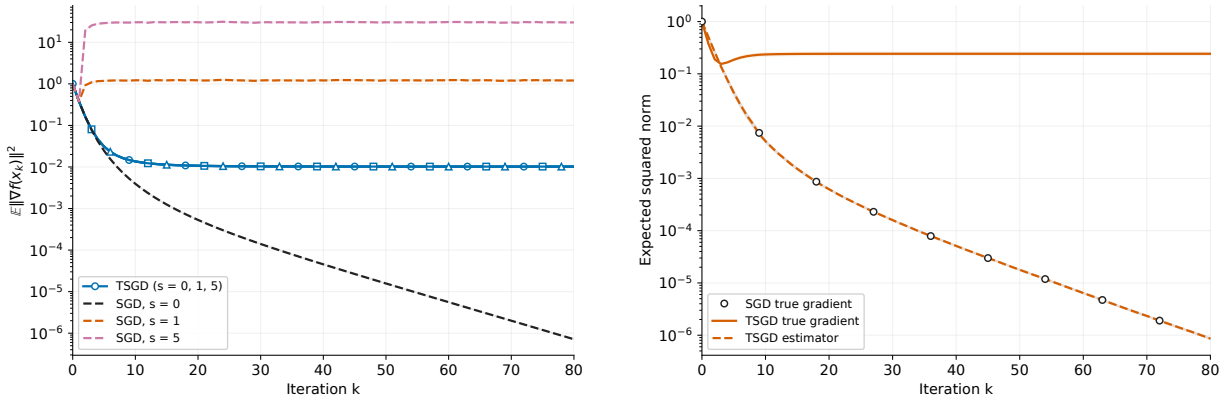
Thus $\delta = 0$, independently of σ. With exact initialization, TSGD follows gradient descent exactly. Indeed, $g_k = \nabla f(x_k)$ and an exact difference give

$$g_{k+1} = g_k + \nabla f(x_{k+1}) - \nabla f(x_k) = \nabla f(x_{k+1}).$$

Induction from $g_0 = \nabla f(x_0)$ proves this for any objective. Appendix B.2 gives the resulting GD convergence guarantee and proves that the gradients tend to zero.

For the present quadratic, take $x_0 = 1$, $\eta = 1$, and any initial estimate $g_0 = 1 + e_0$ with $E_0 = \mathbb{E}[e_0^2] < \infty$, biased or not. Then $x_1 = -e_0$ and $g_1 = g_0 + (x_1 - x_0) = 0$, so TSGD stays at $x_k = -e_0$ for all $k \ge 1$. SGD instead uses a fresh gradient at every update, giving $x_{k+1} = -\xi_k$. Consequently,

$$\boxed{G_k^{\mathrm{TSGD}} = E_0, \qquad G_k^{\mathrm{SGD}} = \sigma^2, \qquad k \ge 1.}$$



(a) Label noise cancels from transport.

(b) False convergence of the transported estimate.

Figure 2: Least-squares experiments with a common stepsize. (a) With exact initialization and matched feature draws, TSGD's trajectories coincide across label-noise levels, while SGD's error floor grows with the noise. (b) With noiseless labels and variable features, SGD's true gradient decays to zero, while TSGD's estimator vanishes but its true gradient does not. Curves show means over 4,000 trajectories with pointwise bands of ± 2 standard errors; the appendix gives the full setup.

Transport preserves the quality of its initial information, good or bad: TSGD has the smaller error when $E_0 < \sigma^2$ and the larger error when $E_0 > \sigma^2$. A single exact gradient evaluation at initialization, $g_0 = \nabla f(x_0)$, gives $E_0 = 0$: TSGD reaches the minimizer in one step and $G_k^{\mathrm{TSGD}} = 0$ for every $k \geq 1$, without any further exact gradient evaluations. In contrast, single-sample initialization $g_0 = h(x_0, \xi_0)$ gives the same error as SGD.

Figure 2(a) illustrates the same separation with varying features. Holding the initial estimate and feature draws fixed, TSGD's trajectories coincide across label-noise levels, while SGD's error floor increases with the label noise. TSGD nevertheless plateaus because $\delta > 0$ and transport errors accumulate at a constant stepsize; without label noise, SGD eventually overtakes it, as the fresh-gradient example in Section 3.3 also illustrates.

3.2 Fixed Noise Magnitude, Variable Similarity

To determine whether noise magnitude alone explains the relative behavior of SGD and TSGD, we compare oracles with the same pointwise noise distribution but different noise variation. For $\omega \geq 0$, consider

$$f(x) = \frac{L}{2}x^2, \qquad h(x, \varphi) = Lx + \sqrt{2}\sigma \cos(\omega x + \varphi), \qquad \varphi \sim \mathrm{Unif}[0, 2\pi). \tag{3.1}$$

Each new sample is an independent phase; a transport difference uses the same phase at both endpoints. For the noise $r(x, \varphi) = h(x, \varphi) - Lx$,

$$\mathbb{E}[r(x, \varphi)^2] = \sigma^2, \qquad \mathbb{E}[|r(x, \varphi) - r(y, \varphi)|^2] = 4\sigma^2 \sin^2\left(\frac{\omega(x - y)}{2}\right). \tag{3.2}$$

The smallest global similarity constant is $\delta = \sigma\omega$ (Appendix D.4). Pointwise noise has the same distribution for every x and ω, so SGD has identical expected squared-gradient dynamics at every frequency. TSGD, however, has exact transport at $\omega = 0$, while its similarity constant grows as $\delta = \sigma\omega$. Its actual transport error depends on the displacement through the sine term.

Simulations in Appendix D.5 (Figure 4) confirm this distinction. At small frequencies, TSGD retains accurate information despite persistent pointwise noise; at $\omega = 2$, its error exceeds SGD's.

Initialization	Exact transport	Stochastic transport
Exact	Gradient descent	Accumulated transport error
Inexact	Persistent initial error	Both errors and their interaction

Table 1: Four regimes in the quadratic model, with a stable constant stepsize. The exact-transport identities also hold for general objectives; the explicit limiting errors and stability condition are established in Appendix B.

Thus, at the same pointwise noise level and stepsize, changing the noise variation changes which method performs better.

3.3 When Fresh Gradients Are Preferable

Now let

$$f(x) = \frac{x^2}{2}, \qquad h(x,a) = ax, \qquad \mathbb{P}(a=0) = \mathbb{P}(a=2) = \frac{1}{2}, \qquad x_0 = 1. \tag{3.3}$$

Here the gradient-noise variance is x^2: fresh gradients become exact at the minimizer, although global BV fails. Similarity holds with $\delta = 1$. At the common stepsize $\eta = 1/2$, each SGD sample either leaves the iterate unchanged or sends it to zero, so $G_k^{\mathrm{SGD}} = 2^{-k}$. In contrast, exactly initialized TSGD satisfies $g_{k+1} = (1 - a_{k+1}/2)g_k$. Its estimator becomes zero at the first sample $a_{k+1} = 2$, wherever the iterate happens to be. The exact calculation in Appendix D.2 gives

$$g_k \longrightarrow 0 \quad \text{almost surely}, \qquad G_k^{\mathrm{TSGD}} \longrightarrow \frac{1}{2}.$$

Rephrase this one, I don't like "need not" here.

This is false convergence: a vanishing estimate need not indicate a vanishing true gradient. Fresh information can therefore be preferable even when initialization is exact.

Figure 2(b) shows the same phenomenon on least squares with noiseless labels and highly variable features. SGD's true gradient decays geometrically, whereas the TSGD estimator decays at nearly the same rate but its true gradient settles near 0.27. A small transported estimate therefore need not certify a small true gradient.

3.4 What Transport Cannot Do

To separate the effects of initialization and transport, consider $f(x) = \lambda x^2/2$ and $h(x;a,b) = (\lambda + a)x + b$, with $\lambda > 0$, independent noises satisfying $\mathbb{E}[a] = \mathbb{E}[b] = 0$, $\mathbb{E}[a^2] = \delta^2$, and $\mathbb{E}[b^2] < \infty$. Shared sampling cancels b. Table 1 summarizes the four cases, whose exact dynamics are developed in Appendix B.

Exact transport preserves initialization error. The behavior in Section 3.1 extends beyond quadratics. When $\delta = 0$, the error identity gives $e_k = e_0$, hence $g_k = \nabla f(x_k) + e_0$ for every k. With $e_0 = 0$ this is GD; otherwise it is GD on $f(x) + \langle e_0, x\rangle$. The obstruction is informational: adding a linear term leaves all gradient differences unchanged, so differences alone cannot recover a missing constant gradient offset (Appendix B.3).

obstruction? Use another word.

Stochastic transport can produce a nonzero error floor. For $0 < \eta < 2\lambda/(\lambda^2 + \delta^2)$ and $M_0 := \mathbb{E}[|g_0|^2]$, the exact quadratic dynamics give

$$\lim_{k\to\infty} \mathbb{E}[|g_k|^2] = 0, \qquad \lim_{k\to\infty} G_k = E_0 + \frac{\eta\delta^2 M_0}{2\lambda - \eta(\lambda^2 + \delta^2)}. \tag{3.4}$$

With exact initialization, $M_0 = \lambda^2 x_0^2$: if $\delta > 0$ and $x_0 \neq 0$, the estimate vanishes in mean square while the true gradient has a positive limiting mean-square norm. Appendix B.4 proves both limits.

The two errors interact. Initialization affects the residual directly through E_0 and indirectly through M_0, which controls the steps and the transport errors they generate. For centered initialization error, $M_0 = \lambda^2 x_0^2 + E_0$, so improving the initial estimate reduces both contributions. Appendix B.5 derives this interaction and the associated conservation law. Appendix B.6 checks the four regimes numerically on least squares and compares the observed residuals with their exact limits.

4 Convergence Theory and Comparison with SGD

The examples distinguish noise magnitude from noise variation. We now make this distinction explicit in finite-horizon stationarity guarantees. Using G_k from the preceding section, write

$$\Delta_0 := f(x_0) - f_{\inf}, \qquad \overline{G}_K := \frac{1}{K} \sum_{k=0}^{K-1} G_k,$$

with a superscript identifying the algorithm when needed.

4.1 Main Convergence Guarantee for TSGD

Using the history $\mathcal{F}_k$ defined in Section 2.3, write $m_k := \nabla f(x_{k+1}) - \nabla f(x_k)$. We impose two conditions on the increments actually used along the trajectory.

Assumption A5 (Unbiased transport)**.** The increments satisfy

$$\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = m_k. \tag{4.1}$$

Assumption A6 (Similarity along the trajectory)**.** For some $\delta \geq 0$,

$$\mathbb{E}[\|D_{k+1} - m_k\|^2 \mid \mathcal{F}_k] \leq \delta^2 \|x_{k+1} - x_k\|^2. \tag{4.2}$$

Assumptions A2 and A4 imply these two conditions for shared-sample gradient differences. The conditional similarity requirement may be weaker than global similarity because it concerns only the generated increments. Stating them directly on the increments also allows transport without a pointwise gradient oracle.

Theorem 4.1 (Finite-horizon stationarity of TSGD)**.** *Suppose the objective is smooth and lower bounded (Assumption A1), and the transport increments are conditionally unbiased and satisfy centered δ-similarity (Assumptions A5–A6). Allow any initial estimate with $E_0 < \infty$, including biased ones. For an integer $K \geq 1$, choose a constant stepsize $\eta > 0$ satisfying*

$$L\eta + (K-1)\delta^2\eta^2 \leq 1. \tag{4.3}$$

Then

$$\overline{G}_K^{\mathrm{TSGD}} \leq \frac{2\Delta_0}{\eta K} + E_0. \tag{4.4}$$

Why the guarantee holds. Let $F_k := \mathbb{E}[f(x_k) - f_{\inf}]$ and $M_k := \mathbb{E}[\|g_k\|^2]$. Smoothness gives

$$F_{k+1} \le F_k - \frac{\eta}{2} G_k - \frac{\eta}{2}(1 - L\eta) M_k + \frac{\eta}{2} E_k. \tag{4.5}$$

The error identity (2.11) and similarity imply $E_k \le E_0 + \delta^2 \eta^2 \sum_{t<k} M_t$. When the descent inequalities are summed, an error generated at step t enters at most $K - 1 - t$ later estimates. Condition (4.3) ensures that its total contribution is no larger than the corresponding negative M_t term. This leaves the initialization error and the initial objective gap in (4.4). Appendix A gives the full proof, retaining the terminal objective gap and all estimator-norm terms. Neither bounded variance nor a pointwise gradient oracle is required.

Choosing the largest admissible stepsize gives

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} \le \frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0.} \tag{4.6}$$

The TSGD bound does not explicitly depend on σ, but this dependence may enter through E_0. For example, initializing g_0 by averaging B_0 independent unbiased stochastic gradients at x_0 gives $E_0 \le \sigma^2/B_0$ under bounded variance. Figure 5 illustrates how a more accurate sampled initial estimate lowers TSGD's residual error. Thus, TSGD's sensitivity to gradient-noise magnitude depends on how its initial estimate is constructed.

The terms correspond to deterministic optimization, accumulated transport noise, and initialization error. Small δ permits larger steps and a smaller transport contribution. With $\delta = 0$, the admissible step is $1/L$; if also $E_0 = 0$, we recover gradient descent. The additive E_0 term reflects the error that pure transport preserves, and its coefficient is sharp (Appendix B). The guarantee approaches E_0 as the horizon grows. This floor can be attained even with exact transport, as Section 3.4 shows. Stochastic transport can create an additional floor when the stepsize is held fixed. For $\delta > 0$, the theorem controls this accumulation by choosing a smaller stepsize for a longer run.

Related single-run guarantees appear in SARAH-IN, iSARAH-IN, and PAGE-NS [17, 18, 21]. The formulation here expresses the argument directly through unbiased increments and their centered similarity.

4.2 Comparison with SGD

For SGD, unbiasedness and BV give the standard smooth nonconvex guarantee [8]. The fixed step $\eta = 1/L$ yields $\overline{G}_K^{\mathrm{SGD}} \le 2L\Delta_0/K + \sigma^2$. For $\Delta_0 > 0$, optimizing the constant step for the prescribed horizon replaces this persistent variance term by a decaying one, as shown in Table 2. The exact stepsizes and derivations are in Appendix A.

The comparison reflects the different information the methods use: SGD's bound depends on noise magnitude, while TSGD's depends on noise variation and initial accuracy. Small δ can favor transport even when σ is large. Neither BV nor the more general ABC condition is comparable with global similarity (Appendix C.1), so these bounds apply under different assumptions and do not establish universal superiority.

4.3 Computational Considerations

TSGD requires an initial estimate and, in the shared pointwise-oracle implementation, two gradient evaluations per transport increment. Accurate differences may nevertheless need far fewer samples than accurate absolute gradients: small-batch transport can preserve information obtained from a more expensive initial estimate. The benefit depends on both its initial cost and the subsequent

	SGD	TSGD
Noise assumption	BV: pointwise magnitude σ	Centered similarity: variation δ
Tuned bound on $\overline{G}_K$	$\frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}$	$\frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0$
Pointwise evaluations	K	$B_0 + 2(K-1)$

Table 2: Finite-horizon guarantees under each method's noise assumptions, with constant stepsizes optimized separately. Both require a smooth, lower-bounded objective. Costs count K iterate updates and an initialization using B_0 evaluations for TSGD.

accuracy of the differences. Iteration counts alone therefore do not compare computational work. Appendix E gives batch initialization costs and comparisons at equal oracle budgets.

5 TSGD as a Building Block of Modern Optimizers

5.1 Transport Is Already Everywhere

Many modern stochastic optimizers combine fresh gradients with transported estimates. Their common building blocks are the three operations in Figure 1: fresh estimation, averaging, and transport. Write H_{k+1} for a fresh unbiased estimate of $\nabla f(x_{k+1})$, D_{k+1} for an estimate of its change from x_k, and $g_k + D_{k+1}$ for the transported estimate. A stochastic gradient difference uses the same fresh sample or minibatch at both endpoints. Table 3 summarizes the resulting estimator relationships. Here $\alpha, \beta \in [0, 1]$ weight fresh estimates, and γ scales the MARS transport correction.

Method	Estimator at iteration $k+1$	Role of transport
SARAH / SPIDER	H_{k+1} or $g_k + D_{k+1}$	Transport between periodic refreshes.
Geom-SARAH	H_{k+1} or $g_k + D_{k+1}$	Transport for a geometric number of steps.
PAGE	H_{k+1} or $g_k + D_{k+1}$	Refresh with probability p.
MARINA	$\nabla f(x_{k+1})$ or $g_k + D^{\mathcal{Q}}_{k+1}$	Communicate compressed gradient changes.
STORM / MVR	$(1-\alpha)(g_k + D_{k+1}) + \alpha H_{k+1}$	Average fresh and transported estimates.
MARS (unclipped)	$(1-\beta)(g_k + \gamma D_{k+1}) + \beta H_{k+1}$	Scale the transport correction within momentum.

Table 3: Transport in modern optimizers. Rows with two alternatives select one branch according to a restart or refresh rule; the last two rows form weighted combinations. $D^{\mathcal{Q}}_{k+1}$ denotes an unbiased compressed increment. The identities describe estimators; batch sizes, stepsizes, and iterate updates belong to each algorithm.

SVRG [12] and SAGA [5] also use stochastic gradient differences to reduce variance. SVRG compares component gradients at the current iterate and a reference point held fixed within each

Is this reference point an anchor point? If so, use that word please.

epoch, adding the full gradient at that reference point. SAGA instead stores past gradients for individual components and combines the sampled component's gradient difference with the average of the stored gradients, updating one entry at each iteration. Both exploit noise cancellation in gradient differences. Unlike TSGD, they do not transport the previous gradient estimate along successive iterates: each estimate combines a current component gradient with stored reference information. In their standard forms, these estimates are conditionally unbiased, whereas pure TSGD inherits and accumulates tracking errors.

SARAH and SPIDER transport an estimate between full-gradient or large-batch recomputations [6, 16, 17]. Inexact SARAH allows batch initialization [18], while Geom-SARAH randomizes the phase length [10]. PAGE refreshes probabilistically [15]; MARINA combines exact refreshes with compressed gradient differences [9].

MVR/STORM instead averages fresh and transported estimates at each iteration [4]. The unclipped MARS estimator [23] includes MSGD at $\gamma = 0$ and MVR at $\gamma = 1$. These identities concern the stated sampling: cached gradients, clipping, and preconditioning require separate treatment. Sampling also affects the noise covariance, as emphasized by the VENUS construction [1]. Appendix E.5 gives the precise identities, error recursions, and special cases.

Gradient differences also appear in decentralized gradient tracking [19], where agents combine changes in their local gradients with information from neighboring agents to track the average gradient across the network.

Neural-network applications. Transport-based methods have been used to train both image classifiers and language models. SARAH and SARAH+ were evaluated on networks with one fully connected hidden layer on MNIST and CIFAR-10 [17]; PAGE was evaluated on LeNet, VGG, and ResNet architectures [15]; and MARS was evaluated on GPT-2 language models [23]. These experiments concern composite optimizers, not uninterrupted pure TSGD: SARAH variants restart their transport phases, PAGE refreshes probabilistically, and MARS combines gradient-difference corrections with momentum and preconditioning. In particular, although the nonconvex SARAH paper analyzes the pure-transport inner loop SARAH-IN, its neural-network experiments use restarted SARAH/SARAH+. Some practical MARS variants also approximate the correction using gradients computed with different samples, rather than the shared-sample increment studied here [23]. These applications motivate studying transport as a building block; they do not establish the performance of pure TSGD in isolation.

5.2 Why Combine Fundamental Estimators?

SGD, MSGD, and TSGD use fresh information, averaging, and transport, respectively. Modern optimizers combine them through weighted averages or alternate between fresh estimation and transport. Three motivations explain why these combinations can be useful.

Computational efficiency. An accurate absolute gradient may require a large batch, whereas a small batch can suffice for an accurate gradient change. Even with two evaluations per sample, transport can then cost much less. SARAH, SPIDER, and PAGE preserve information from an expensive gradient estimate through many cheaper transport steps. Refreshing more often limits accumulated error but reduces this saving.

Here you should say that momentum's estimate can lag behind, especially when the gradient changes fast.

Complementary strengths. Momentum averages noisy information from previous iterations, but those gradients were evaluated at points the iterate has already left. Transport accounts for the change in the true gradient, reducing tracking lag while retaining the benefit of averaging. Fresh gradients supply absolute information that pure transport cannot recover: mixing them in damps initialization and accumulated transport errors, while introducing new pointwise noise. MVR,

STORM, and MARS balance these roles through their mixing weights and transport corrections. This complementarity matters even when the noise regime remains fixed.

Changing regimes. The relative accuracy of fresh and transported estimates can change during a single run. Similarity bounds each new transport error in proportion to the squared displacement, so shorter steps tighten this bound without removing previously accumulated error. Section 3.2 illustrates this displacement dependence even when pointwise noise variance stays constant. Fresh gradients may also become more reliable: in Section 3.3, their noise vanishes near the minimizer, while pure transport can retain a nonzero tracking error. Thus the balance can change as iterates move, steps shrink, and errors accumulate.

Mixing weights and refresh frequency determine how much an optimizer relies on each mechanism. Fixed choices provide a compromise, while adapting them can respond to changing conditions. The balance matters: combining estimators does not automatically improve convergence, and a poor choice can perform worse than using one estimator alone.

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Improved complexity through refreshing. When bounded variance and centered similarity hold together, combining estimators can improve the available convergence guarantees on the same problem class. SGD repeatedly obtains fresh gradient information but does not explicitly exploit accurate shared-sample gradient differences. Pure TSGD exploits these differences, but obtains absolute-gradient information only through initialization. It retains initialization error and accumulates transport noise.

SPIDER [6] and PAGE [15] combine transport with periodic or probabilistic refreshing. An accurate gradient estimate can then support many inexpensive transport steps, while each refresh replaces the inherited error with the error of a fresh estimate. This limits how long transport errors accumulate and allows larger steps without requiring an expensive gradient estimate at every iteration. With suitable batch sizes and refresh schedules, these methods attain $O(\varepsilon^{-3})$ oracle complexity, compared with the $O(\varepsilon^{-4})$ guarantees for SGD and pure TSGD in Section 4. This ε^{-3} dependence matches the lower bound of Fradin et al. [7] for zero-respecting algorithms on this same problem class; Appendix E.4 gives the precise statement. The comparison shows how combining the mechanisms can improve a guarantee; it does not establish an $\Omega(\varepsilon^{-4})$ lower bound for pure TSGD. Pure TSGD can already converge faster when transport is exact.

I don't see the point of that sentence.

6 Conclusion

Transported SGD is a fundamental way to use stochastic-gradient information: it estimates changes and carries an existing estimate along the trajectory. Its strength becomes visible when noise is understood through its variation as well as its magnitude. Accurate initialization and high similarity can make transport effective even when individual gradients are very noisy. Its persistent initialization error and accumulated transport errors explain both its limitations and its complementarity with fresh estimation and momentum averaging. Recognizing these distinct mechanisms clarifies why transport appears throughout modern optimization, and why the balance between them matters.

horrible phrasing.

In the conclusion, also mention that we believed it is worth it to study this fundamental mechanism on its own.

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A Detailed Convergence Proofs

A.1 Descent and Weighted Accumulation

Fix deterministic x_0, a horizon $K \geq 1$, and a constant step $\eta > 0$. In addition to the notation of Section 4, write $q_k := \mathbb{E}[\|d_{k+1}\|^2]$. The finite second moment of g_0, smoothness, and conditional similarity imply by induction that x_k, g_k, e_k have finite second moments at every finite k. Smoothness also bounds the growth of f by a quadratic, so the expectations below are well defined. The assumption $L > 0$ excludes only a trivial case: a lower-bounded function with 0-Lipschitz gradient is constant.

For completeness, e_k is $\mathcal{F}_k$-measurable and $\mathbb{E}[d_{k+1} \mid \mathcal{F}_k] = 0$. Therefore

$$\mathbb{E}[\langle e_k, d_{k+1}\rangle] = \mathbb{E}[\langle e_k, \mathbb{E}[d_{k+1} \mid \mathcal{F}_k]\rangle] = 0, \qquad E_{k+1} = E_k + q_k.$$

Iteration yields (2.11); more generally, distinct transport errors are orthogonal in expectation. The process (e_k) is a square-integrable martingale for the history containing the initialization and all transport samples observed by step k.

The smoothness inequality and the identity

$$2\langle \nabla f(x_k), g_k\rangle = \|\nabla f(x_k)\|^2 + \|g_k\|^2 - \|e_k\|^2$$

give

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \eta\langle \nabla f(x_k), g_k\rangle + \frac{L\eta^2}{2}\|g_k\|^2 \\ &= f(x_k) - \frac{\eta}{2}\|\nabla f(x_k)\|^2 - \frac{\eta}{2}(1 - L\eta)\|g_k\|^2 + \frac{\eta}{2}\|e_k\|^2. \end{aligned}$$

Taking expectations proves (4.5). The martingale identity (2.11) yields

$$\sum_{k=0}^{K-1} E_k = KE_0 + \sum_{k=0}^{K-1}\sum_{t=0}^{k-1} q_t = KE_0 + \sum_{t=0}^{K-2}(K - 1 - t)q_t. \tag{A.1}$$

The last sum is empty for $K = 1$. Similarity gives

$$q_t \leq \delta^2\mathbb{E}[\|x_{t+1} - x_t\|^2] = \delta^2\eta^2 M_t. \tag{A.2}$$

The weight $K - 1 - t$ counts the later estimates that inherit the error generated by the tth displacement.

A.2 Proof of Theorem 4.1

Summing (4.5) and substituting (A.1) gives

$$F_K + \frac{\eta}{2}\sum_{k=0}^{K-1} G_k + \frac{\eta}{2}(1 - L\eta)\sum_{k=0}^{K-1} M_k \leq \Delta_0 + \frac{\eta K}{2}E_0 + \frac{\eta}{2}\sum_{t=0}^{K-2}(K - 1 - t)q_t.$$

Apply (A.2) and collect the coefficients of M_k. For $c_k := 1 - L\eta - \delta^2\eta^2(K - 1 - k)$, this proves the stronger bound

$$\frac{1}{K}\sum_{k=0}^{K-1} G_k + \frac{2F_K}{\eta K} + \frac{1}{K}\sum_{k=0}^{K-1} c_k M_k \leq \frac{2\Delta_0}{\eta K} + E_0. \tag{A.3}$$

Condition (4.3) ensures $c_k \geq 0$ for every $0 \leq k < K$. Since also $F_K \geq 0$, dropping the last two terms on the left proves (4.4).

A.3 Exact Optimized TSGD Guarantee

The largest positive step satisfying $L\eta + (K-1)\delta^2\eta^2 \le 1$ is

$$\boxed{\eta^\star = \frac{2}{L + \sqrt{L^2 + 4(K-1)\delta^2}}.} \tag{A.4}$$

For $(K-1)\delta^2 > 0$, this is the positive root of the quadratic, written after rationalizing its numerator. The same expression gives $1/L$ when $K = 1$ or $\delta = 0$. It maximizes the step under the theorem's sufficient condition; it is not a necessary stability limit. Substitution into (4.4) yields

$$\overline{G}_K^{\mathrm{TSGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + 4(K-1)\delta^2}\right) + E_0. \tag{A.5}$$

Using

$$\sqrt{L^2 + 4(K-1)\delta^2} \le L + 2\delta\sqrt{K-1}, \qquad \frac{\sqrt{K-1}}{K} \le \frac{1}{\sqrt{K}},$$

gives (4.6). For $(K-1)\delta^2 \ll L^2$, $\eta^\star$ is close to $1/L$; for $(K-1)\delta^2 \gg L^2$, it is of order $1/(\delta\sqrt{K})$. These are the deterministic and transport-dominated regimes of this bound. Exact initialization removes E_0 but does not remove the transport term; exact transport removes the latter but preserves E_0.

A.4 SGD under Bounded Variance

Under Assumptions A1–A3, let F_k, G_k now refer to the SGD iterates. Conditional unbiasedness and BV imply

$$\mathbb{E}_\xi[\|h(x_k,\xi)\|^2 \mid x_k] \le \|\nabla f(x_k)\|^2 + \sigma^2.$$

Consequently,

$$F_{k+1} \le F_k - \eta\left(1 - \frac{L\eta}{2}\right)G_k + \frac{L\eta^2\sigma^2}{2}.$$

Summing and dropping $F_K \ge 0$ yields, for $0 < \eta < 2/L$,

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{2\Delta_0 + KL\eta^2\sigma^2}{\eta K(2 - L\eta)}.} \tag{A.6}$$

In particular, the fixed gradient-descent step $\eta = 1/L$ gives

$$\overline{G}_K^{\mathrm{SGD}} \le \frac{2L\Delta_0}{K} + \sigma^2. \tag{A.7}$$

For $\Delta_0 > 0$, let $U(\eta) := (2\Delta_0/K + L\sigma^2\eta^2)/(\eta(2 - L\eta))$. Its derivative has the sign of

$$L\sigma^2\eta^2 + \frac{2L\Delta_0}{K}\eta - \frac{2\Delta_0}{K}.$$

This expression is strictly increasing for $\eta > 0$ and changes sign once in $(0, 2/L)$. Thus the minimizing constant step is

$$\boxed{\eta^*_{\mathrm{BV},K} = \frac{2}{L + \sqrt{L^2 + 2KL\sigma^2/\Delta_0}}.} \tag{A.8}$$

At the root, $L\sigma^2\eta^2 = (2\Delta_0/K)(1 - L\eta)$, so $U(\eta) = 2\Delta_0/(K\eta)$. Hence

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}}\right) \le \frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}.} \tag{A.9}$$

The last inequality uses $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. When $\sigma^2 = 0$, the step is $1/L$ and the bound reduces to $2L\Delta_0/K$. When $\sigma^2 > 0$, the step decreases with K, giving a finite-horizon guarantee rather than an assertion about an infinite run with a fixed positive step. If $\Delta_0 = 0$, x_0 is already stationary; the optimized formula above is not used.

A.5 Iteration Complexity

Fix an accuracy target $\varepsilon > 0$ and let R be uniform on $\{0, \dots, K-1\}$, independent of the run. Then $\mathbb{E}[\|\nabla f(x_R)\|^2] = \overline{G}_K$.

Corollary A.1 (Sufficient TSGD horizon)**.** *Under Theorem 4.1, suppose $E_0 \le \varepsilon^2/3$ and use $\eta = \eta^\star$. An integer horizon satisfying*

$$K \ge \max\left\{1, \frac{6L\Delta_0}{\varepsilon^2}, \frac{36\delta^2\Delta_0^2}{\varepsilon^4}\right\} \tag{A.10}$$

ensures $\mathbb{E}[\|\nabla f(x_R)\|^2] \le \varepsilon^2$.

Proof. Each of the first two terms in (4.6) is at most $\varepsilon^2/3$ under (A.10); the initialization term satisfies the same bound by assumption. □

For SGD with $\Delta_0 > 0$ and step (A.8), the analogous sufficient horizon is

$$K \ge \max\left\{1, \frac{4L\Delta_0}{\varepsilon^2}, \frac{8L\Delta_0\sigma^2}{\varepsilon^4}\right\}. \tag{A.11}$$

Each of the two terms in the simplified SGD bound is then at most $\varepsilon^2/2$. These are iteration upper bounds; initialization and per-iteration oracle costs are accounted for in Appendix E.

B Exact Quadratic Dynamics and Error Sources

B.1 The Quadratic Model

Let x_0 be deterministic, $\eta > 0$ constant, and

$$f(x) = \frac{\lambda}{2}x^2, \qquad h(x; a, b) = (\lambda + a)x + b, \qquad \lambda > 0. \tag{B.1}$$

The noises are independent and satisfy

$$\mathbb{E}[a] = \mathbb{E}[b] = 0, \qquad \mathbb{E}[a^2] = \delta^2, \qquad \mathbb{E}[b^2] = \sigma^2.$$

Thus $\nabla f(x) = \lambda x$ and $\mathbb{E}[h(x; a, b)] = \nabla f(x)$. Each transport step uses a fresh pair (a_{k+1}, b_{k+1}), independent of the past and the square-integrable initial estimate g_0, at both endpoints. The additive noise cancels:

$$D_{k+1} = (\lambda + a_{k+1})(x_{k+1} - x_k). \tag{B.2}$$

Consequently,

$$\mathbb{E}[|D_{k+1} - [\nabla f(x_{k+1}) - \nabla f(x_k)]|^2 \mid \mathcal{F}_k] = \delta^2 |x_{k+1} - x_k|^2.$$

Assumption A6 holds with equality at $\delta = \sqrt{\mathbb{E}[a^2]}$, independently of σ. Substituting $x_{k+1} - x_k = -\eta g_k$ in $g_{k+1} = g_k + D_{k+1}$ gives

$$g_{k+1} = (1 - \eta\lambda - \eta a_{k+1}) g_k, \qquad e_{k+1} = e_k - \eta a_{k+1} g_k. \tag{B.3}$$

We retain $E_k = \mathbb{E}[|e_k|^2]$, $M_k = \mathbb{E}[|g_k|^2]$, and $G_k = \mathbb{E}[|\nabla f(x_k)|^2]$. In particular, tracking error E_k and true stationarity error G_k are distinct quantities.

B.2 Exact Initialization and Exact Transport

For any objective, if $g_0 = \nabla f(x_0)$ and $D_{k+1} = \nabla f(x_{k+1}) - \nabla f(x_k)$, induction gives

$$g_{k+1} = g_k + D_{k+1} = \nabla f(x_{k+1}), \qquad x_{k+1} = x_k - \eta \nabla f(x_k).$$

TSGD is exactly GD, path by path. Under Assumption A1, taking its standard stepsize $\eta = 1/L$ in Theorem 4.1 gives

$$\frac{1}{K} \sum_{k=0}^{K-1} \|\nabla f(x_k)\|^2 \le \frac{2L\Delta_0}{K}. \tag{B.4}$$

Moreover, GD descent yields, for every K,

$$\frac{1}{2L} \sum_{k=0}^{K-1} \|\nabla f(x_k)\|^2 \le f(x_0) - f(x_K) \le \Delta_0.$$

The nonnegative partial sums are uniformly bounded. Thus $\sum_{k\ge 0} \|\nabla f(x_k)\|^2 < \infty$ and $\|\nabla f(x_k)\|^2 \to 0$, without convexity or convergence of the iterates themselves.

In (B.1), $\delta = 0$ means $a = 0$ almost surely, so transport is exact despite any additive noise. With $g_0 = \lambda x_0 = \nabla f(x_0)$ and $r := 1 - \eta\lambda$,

$$x_k = r^k x_0, \qquad g_k = \lambda r^k x_0, \qquad E_k = 0. \tag{B.5}$$

For $0 < \eta < 2/\lambda$, $|r| < 1$ and $G_k = \lambda^2 r^{2k} x_0^2 \to 0$.

B.3 Inexact Initialization and Exact Transport

For any objective, let $g_0 = \nabla f(x_0) + e_0$ and take exact differences. Then $e_k = e_0$ for every k, and

$$g_k = \nabla f(x_k) + e_0 = \nabla \widetilde{f}_{e_0}(x_k), \qquad \widetilde{f}_{e_0}(x) := f(x) + \langle e_0, x \rangle.$$

Hence TSGD is GD on a fixed linearly tilted objective, and $E_k = E_0$. At $\eta = 1/L$, Theorem 4.1 gives

$$\frac{1}{K} \sum_{k=0}^{K-1} G_k \le \frac{2L\Delta_0}{K} + E_0. \tag{B.6}$$

For fixed e_0, if $\widetilde{f}_{e_0}$ is also bounded below, GD descent on that objective implies $\nabla f(x_k) \to -e_0$ and $\|\nabla f(x_k)\|^2 \to \|e_0\|^2$. This additional lower bound is required for the pathwise asymptotic claim, not for (B.6).

The informational obstruction is that

$$f_c(x) := f(x) + \langle c, x\rangle \quad \Longrightarrow \quad \nabla f_c(x) - \nabla f_c(y) = \nabla f(x) - \nabla f(y).$$

If TSGD is applied to f_c but initialized with $\nabla f(x_0)$, exact transport follows GD on f, so missing absolute-gradient information cannot be recovered from differences alone.

For the quadratic and $\delta = 0$, the exact dynamics are

$$g_k = r^k(\lambda x_0 + e_0), \qquad x_k = r^k x_0 - \frac{1 - r^k}{\lambda} e_0. \tag{B.7}$$

The first identity follows from (B.3); the second follows from $\lambda x_k = g_k - e_0$. If $\mathbb{E}[e_0] = 0$, then

$$G_k = r^{2k}\lambda^2 x_0^2 + (1 - r^k)^2 E_0. \tag{B.8}$$

For $0 < \eta < 2/\lambda$, even with biased square-integrable e_0, $x_k \to -e_0/\lambda$ in mean square and

$$\lim_{k\to\infty} G_k = E_0. \tag{B.9}$$

For centered additive initialization $e_0 = b_0$ of variance σ^2, this gives $\mathbb{E}[x_\infty] = 0$ but $\mathrm{Var}(x_\infty) = \sigma^2/\lambda^2$ and $\lim_k G_k = \sigma^2$: a correct mean iterate can conceal random nonoptimal limits.

Sharpness of the initialization coefficient. Choose an initialization with $E_0 > 0$. For $\varepsilon > 0$, choose N with $|G_k - E_0| \le \varepsilon$ for $k \ge N$. For every $K > N$,

$$\left|\frac{1}{K}\sum_{k=0}^{K-1} G_k - E_0\right| \le \frac{1}{K}\sum_{k=0}^{N-1} |G_k - E_0| + \varepsilon.$$

Letting $K \to \infty$ and then $\varepsilon \downarrow 0$ gives, at the admissible $\eta = 1/L = 1/\lambda$,

$$\lim_{K\to\infty} \frac{1}{K}\sum_{k=0}^{K-1} G_k = E_0 = \lim_{K\to\infty}\left(\frac{2\Delta_0}{\eta K} + E_0\right).$$

Thus the coefficient one on E_0 in (4.4) is sharp.

B.4 Exact Initialization and Stochastic Transport

Take $g_0 = \lambda x_0$ and $\delta > 0$, and define

$$r := 1 - \eta\lambda, \qquad \rho := \mathbb{E}[(r - \eta a)^2] = r^2 + \eta^2\delta^2 = 1 - 2\eta\lambda + \eta^2(\lambda^2 + \delta^2).$$

Independence and centeredness of a_{k+1} give

$$M_{k+1} = \rho M_k, \qquad E_{k+1} = E_k + \eta^2\delta^2 M_k. \tag{B.10}$$

Starting from $E_0 = 0$ and $M_0 = \lambda^2 x_0^2$,

$$M_k = \rho^k \lambda^2 x_0^2, \qquad E_k = \eta^2\delta^2\lambda^2 x_0^2 \sum_{t=0}^{k-1} \rho^t. \tag{B.11}$$

The stability condition is

$$\rho < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{2\lambda}{\lambda^2 + \delta^2}. \tag{B.12}$$

Under this condition, $g_k \to 0$ in mean square, while

$$E_\infty := \lim_{k\to\infty} E_k = \frac{\eta\delta^2\lambda^2 x_0^2}{2\lambda - \eta(\lambda^2+\delta^2)}. \tag{B.13}$$

To relate this tracking error to the true gradient, note that $e_{t+1} - e_t = -\eta a_{t+1} g_t$ are orthogonal martingale increments. For $m > n$,

$$\mathbb{E}[|e_m - e_n|^2] = \eta^2\delta^2 \sum_{t=n}^{m-1} M_t \le \frac{\eta^2\delta^2 M_0}{1-\rho}\rho^n \longrightarrow 0.$$

Thus (e_k) is Cauchy in the complete space L^2, and converges in mean square to some e_∞. Since $g_k \to 0$ in mean square, $x_k = (g_k - e_k)/\lambda \to -e_\infty/\lambda =: x_\infty$ in mean square as well. Consequently,

$$\lim_{k\to\infty} M_k = 0, \qquad \lim_{k\to\infty} G_k = \mathbb{E}[|\nabla f(x_\infty)|^2] = \mathbb{E}[|e_\infty|^2] = E_\infty > 0 \quad (x_0 \neq 0).$$

The estimator converges to zero in mean square while the true gradient does not. This is the false-convergence phenomenon.

B.5 Inexact Initialization and Stochastic Transport

Allow arbitrary square-integrable e_0 and retain (B.12). The moment recursions remain valid. Define

$$A := \frac{\eta^2\delta^2}{1-\rho} = \frac{\eta\delta^2}{2\lambda - \eta(\lambda^2+\delta^2)}. \tag{B.14}$$

Since $A(1-\rho) = \eta^2\delta^2$,

$$E_{k+1} + AM_{k+1} = E_k + (\eta^2\delta^2 + A\rho)M_k = E_k + AM_k.$$

Induction proves the conservation law

$$E_k + AM_k = E_0 + AM_0 \qquad (k \ge 0). \tag{B.15}$$

Since $M_k = \rho^k M_0 \to 0$, the exact interaction formula follows:

$$\boxed{E_\infty = E_0 + \frac{\eta\delta^2}{2\lambda - \eta(\lambda^2+\delta^2)} M_0 = E_0 + AM_0.} \tag{B.16}$$

The preceding L^2 argument also gives $G_k \to E_\infty$. Initialization contributes directly through E_0 and indirectly through the initial estimator magnitude

$$M_0 = \lambda^2 x_0^2 + 2\lambda x_0 \mathbb{E}[e_0] + E_0.$$

Bias can therefore increase or decrease the transport contribution. For $g_0 = \lambda x_0 + b_0$, with centered b_0 independent of future samples and $\mathbb{E}[b_0^2] = \sigma^2$, the three contributions are

$$E_\infty = \underbrace{\sigma^2}_{\text{initialization}} + \underbrace{A\lambda^2 x_0^2}_{\text{transport}} + \underbrace{A\sigma^2}_{\text{interaction}}. \tag{B.17}$$

In Table 1, the four corresponding limiting stationarity errors are respectively 0, $A\lambda^2 x_0^2$, E_0, and $E_0 + AM_0$. The exact-transport identities are general; these stochastic-transport formulas rely on the quadratic model.

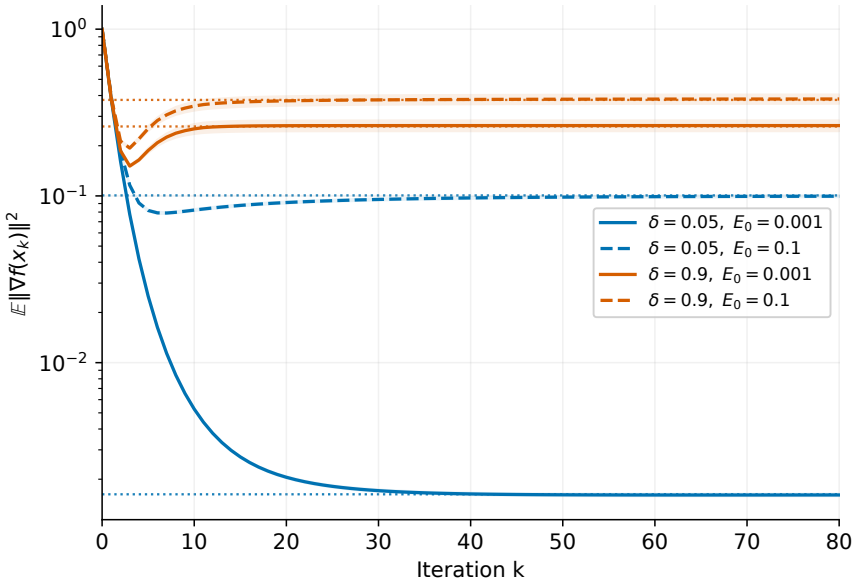


Figure 3: Initialization and transport errors on vector least squares. Each curve uses one of $\delta \in \{0.05, 0.9\}$ and $E_0 \in \{10^{-3}, 10^{-1}\}$; dotted lines are the exact limiting stationarity errors in (B.18). Means over 4,000 trajectories with pointwise bands of ± 2 standard errors.

B.6 Numerical Check on Least Squares

The least-squares experiments in Appendix D.6 use a diagonal quadratic with eigenvalues λ_j. In each coordinate, the shared feature multiplier gives the quadratic recursion above with $a_{j,k} = (v_k - 1)\lambda_j$; the additive label-noise term cancels from every transport increment. Thus each coordinate has the same moment recursions as in (B.10), with $\delta_j^2 = \kappa^2\lambda_j^2$.

For a centered initialization error with $\mathbb{E}[e_{0,j}^2] = E_0/d$, define

$$\rho_j = (1 - \eta\lambda_j)^2 + \eta^2\kappa^2\lambda_j^2, \qquad A_j = \frac{\eta\kappa^2\lambda_j}{2 - \eta(1+\kappa^2)\lambda_j}.$$

When $\rho_j < 1$, the coordinatewise result (B.16) gives

$$\lim_{k\to\infty} \mathbb{E}\|\nabla f(x_k)\|^2 = E_0 + \sum_{j=1}^{d} A_j \left(\lambda_j^2 x_{0,j}^2 + \frac{E_0}{d}\right). \tag{B.18}$$

The figure checks these limits for $d = 10$, $\eta = 1/2$, $\lambda_j \in \{0.1, 0.2, \ldots, 1\}$, and $\kappa \in \{0.05, 0.9\}$. Since $L = \max_j \lambda_j = 1$, the global centered-similarity constant is $\delta = \kappa$. The dotted lines in Figure 3 show the exact limits in (B.18).

C Noise Assumptions

C.1 Bounded Variance, ABC, and Centered Similarity

Write $h(x,\xi) = \nabla f(x) + r(x,\xi)$ with $\mathbb{E}_\xi[r(x,\xi)] = 0$.

ABC generalizes BV. BV requires the gradient-noise variance to be uniformly bounded over the domain. The more general ABC condition, introduced as expected smoothness by Khaled and Richtárik [14], assumes that some $A, B, C \geq 0$ satisfy

$$\mathbb{E}_\xi[\|h(x,\xi)\|^2] \leq 2A(f(x) - f_{\inf}) + B\|\nabla f(x)\|^2 + C \qquad \text{for all } x. \tag{C.1}$$

Unbiasedness gives

$$\mathbb{E}_\xi[\|h(x,\xi)\|^2] = \|\nabla f(x)\|^2 + \mathbb{E}_\xi[\|r(x,\xi)\|^2].$$

BV is the special case $(A, B, C) = (0, 1, \sigma^2)$. ABC permits $\mathbb{E}_\xi[\|h(x,\xi)\|^2]$ to grow with $f(x) - f_{\inf}$ or $\|\nabla f(x)\|^2$. In particular, if $C = 0$ and a sequence (x_k) satisfies

$$f(x_k) - f_{\inf} \to 0 \qquad \text{and} \qquad \|\nabla f(x_k)\|^2 \to 0,$$

then ABC implies

$$\mathbb{E}_\xi[\|h(x_k,\xi)\|^2] \to 0.$$

Thus the second-moment term in SGD's descent bound can vanish instead of being controlled by a fixed noise bound.

For example, let $f(x) = x^2/2$ and $h(x,a) = ax$, where a equals 0 or 2 with equal probability. Its residual is $r(x,a) = h(x,a) - \nabla f(x) = (a-1)x$, so

$$\mathbb{E}[h(x,a)] = x, \qquad \mathbb{E}[|h(x,a)|^2] = 2x^2, \qquad \mathbb{E}[|r(x,a)|^2] = x^2.$$

Thus ABC holds with $(A, B, C) = (0, 2, 0)$, while global BV fails because $\mathbb{E}[|h(x,a) - \nabla f(x)|^2] = \mathbb{E}[|r(x,a)|^2] = x^2$ is unbounded as $|x| \to \infty$. As in Section 3.3, fresh gradients become exact at the minimizer despite unbounded noise variance over the domain.

One-point and two-point conditions. BV and ABC constrain the stochastic gradient at each point separately. Centered similarity instead constrains noise differences under shared sampling, and therefore depends on how the oracle couples two points. In Section 3.2, the pointwise gradient distributions are unchanged as the frequency ω varies, while the smallest similarity constant is $\delta = \sigma\omega$. Thus the same pointwise distributions can coexist with arbitrarily different transport accuracy.

The counterexamples below concern global oracle conditions and show that neither BV nor ABC implies centered similarity, while centered similarity implies neither BV nor ABC.

Neither BV nor ABC implies global similarity. Take $f(x) = x^2/2$ and $h(x,\xi) = x + \xi\sin(x^2)$, where $\mathbb{P}(\xi = -1) = \mathbb{P}(\xi = 1) = 1/2$. The oracle is unbiased and

$$\mathbb{E}_\xi[|h(x,\xi) - \nabla f(x)|^2] = \sin^2(x^2) \leq 1,$$

so BV holds with $\sigma^2 = 1$, and ABC holds with $(A, B, C) = (0, 1, 1)$. Under shared sampling,

$$\mathbb{E}_\xi[|r(x,\xi) - r(y,\xi)|^2] = \big(\sin(x^2) - \sin(y^2)\big)^2.$$

A finite similarity constant would imply

$$|\sin(x^2) - \sin(y^2)| \le \delta |x - y| \qquad \text{for every } x, y,$$
$$|2x\cos(x^2)| = \lim_{y \to x} \frac{|\sin(y^2) - \sin(x^2)|}{|y - x|} \le \delta \quad \text{for every } x.$$

However, at $x_n = \sqrt{2\pi n}$, $n \ge 1$, this would require

$$|2x_n \cos(x_n^2)| = 2\sqrt{2\pi n} \le \delta \qquad \text{for every } n.$$

For any finite δ, choosing an integer $n > \delta^2/(8\pi)$ violates this inequality. Hence no finite global similarity constant exists.

Global similarity implies neither BV nor ABC. Take $f(x) = \log\cosh x$ and $h(x, \xi) = \tanh x + \xi x$, with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \delta^2 > 0$. The objective is lower bounded and 1-smooth, since

$$f(x) \ge 0, \qquad \nabla f(x) = \tanh x, \qquad f''(x) = \frac{1}{\cosh^2 x} \in (0, 1].$$

The unbiased oracle has residual $r(x, \xi) = \xi x$, and hence

$$\mathbb{E}_\xi[|r(x, \xi) - r(y, \xi)|^2] = \delta^2 (x - y)^2, \qquad \mathbb{E}_\xi[|r(x, \xi)|^2] = \delta^2 x^2.$$

Similarity holds with equality, but no finite global BV constant exists. Since $f_{\inf} = 0$, ABC would require

$$\tanh^2 x + \delta^2 x^2 \le 2A \log\cosh x + B \tanh^2 x + C \le 2A|x| + B + C \qquad \text{for all } x.$$

For $x \ne 0$, dividing by x^2 and dropping the nonnegative term $\tanh^2 x / x^2$ gives

$$\delta^2 \le \frac{2A}{|x|} + \frac{B + C}{x^2}.$$

For any finite positive A, B, C, the right-hand side tends to zero as $|x| \to \infty$, whereas the left-hand side is the fixed positive constant δ^2. The inequality therefore fails for sufficiently large $|x|$, so no finite ABC constants exist.

C.2 A TSGD Guarantee without Bounded Variance

For the log-cosh example above, $f_{\inf} = 0$ and the fresh coupled increment is

$$D_{k+1} = \tanh x_{k+1} - \tanh x_k + \xi_{k+1}(x_{k+1} - x_k).$$

Its conditional error variance is exactly $\delta^2 (x_{k+1} - x_k)^2$. Therefore, for deterministic x_0 and arbitrary initialization with finite E_0, (A.5) gives

$$\overline{G}_K^{\mathrm{TSGD}} \le \frac{\log\cosh x_0}{K} \left(1 + \sqrt{1 + 4(K-1)\delta^2}\right) + E_0. \tag{C.2}$$

With exact initialization, this is an $O(K^{-1/2})$ stationarity guarantee even though pointwise gradient variance is unbounded.

For comparison, fixed-step SGD satisfies $x_{k+1} = x_k - \eta \tanh x_k - \eta\xi_k x_k$. For $V_k := \mathbb{E}[x_k^2]$, independence and centering give

$$\begin{aligned} V_{k+1} &= \mathbb{E}[(x_k - \eta \tanh x_k)^2] + \eta^2\delta^2 V_k \\ &\geq (1 + \eta^2\delta^2)V_k - 2\eta\sqrt{V_k}, \end{aligned}$$

where $|\tanh x| \leq 1$ and Cauchy–Schwarz were used. If $|x_0| \geq 4/(\eta\delta^2)$, then

$$\sqrt{V_k} \geq \frac{4}{\eta\delta^2} \quad \Longrightarrow \quad 2\eta\sqrt{V_k} \leq \frac{\eta^2\delta^2}{2} V_k \quad \Longrightarrow \quad V_{k+1} \geq \left(1 + \frac{\eta^2\delta^2}{2}\right) V_k.$$

Induction preserves the threshold and proves divergence of the iterate second moment. This does not establish failure of stationarity, which concerns $\mathbb{E}[\tanh^2 x_k]$. Failure of BV alone is not evidence that SGD cannot converge.

C.3 Quadratic Finite Sums and the Smallest Similarity Constant

For $f_i(x) = \frac{1}{2}x^\top A_i x + b_i^\top x + c_i$ with symmetric A_i, let $f = n^{-1}\sum_i f_i$, $\bar{A} = n^{-1}\sum_i A_i$, and sample i uniformly. By (2.8), the centered difference is $(A_i - \bar{A})(x - y)$. For $z = x - y$,

$$\mathbb{E}_i[\|(A_i - \bar{A})z\|^2] = z^\top \left(\frac{1}{n}\sum_{i=1}^n (A_i - \bar{A})^2\right) z.$$

Taking the supremum over $\|z\| = 1$ gives the smallest admissible global constant,

$$\delta_{\min}^2 = \lambda_{\max}\left(\frac{1}{n}\sum_{i=1}^n (A_i - \bar{A})^2\right). \tag{C.3}$$

The linear terms cancel irrespective of their magnitude, while the smoothness constant of the average quadratic is $L = \|\bar{A}\|_{\mathrm{op}}$. There is no fixed ordering between δ and L. Identical A_i give $\delta_{\min} = 0$ even with large L and arbitrarily different b_i. Conversely, for $n \geq 3$ scalar components, take $f_1(x) = nLx^2/2$ and $f_i(x) = 0$ for $i \geq 2$. Then $f(x) = Lx^2/2$ but

$$\delta_{\min}^2 = \frac{1}{n}((nL - L)^2 + (n-1)L^2) = (n-1)L^2 > L^2.$$

Thus similarity measures the discrepancy between stochastic and deterministic gradient changes, rather than the smoothness of their average.

D Detailed Illustrative Examples

D.1 Additive Noise: Exact Trajectories and Horizon Tuning

For explicit constants, specialize the additive example to

$$f(x) = \frac{x^2}{2}, \qquad h(x,\xi) = x + \xi, \qquad \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = \frac{1}{2}, \qquad x_0 = 1. \tag{D.1}$$

Here $L = 1$, $\Delta_0 = 1/2$, $\sigma^2 = 1$, and $\delta = 0$. At $\eta = 1$, SGD satisfies $x_{k+1} = -\xi_k$, hence

$$G_k^{\mathrm{SGD}} = 1 \quad (k \geq 0), \qquad \boxed{\overline{G}_K^{\mathrm{SGD}} = 1.} \tag{D.2}$$

Thus the fixed-step noise floor is present in the actual trajectory.

For the horizon-dependent BV stepsize (A.8), write $\eta_K = 2/(1+\sqrt{1+4K})$ and $r_K = 1-\eta_K$. Independence and centeredness of the fresh sample give

$$G_{k+1}^{\mathrm{SGD}} = \mathbb{E}[(r_K x_k - \eta_K \xi_k)^2] = r_K^2 G_k^{\mathrm{SGD}} + \eta_K^2.$$

Solving from $G_0 = 1$ and averaging the geometric terms yields

$$G_k^{\mathrm{SGD}} = r_K^{2k} + \frac{\eta_K}{2-\eta_K}(1 - r_K^{2k}),$$
$$\overline{G}_K^{\mathrm{SGD}} = \frac{\eta_K}{2-\eta_K} + \left(1 - \frac{\eta_K}{2-\eta_K}\right)\frac{1-r_K^{2K}}{K(1-r_K^2)}.$$

Since $\eta_K \sim K^{-1/2}$, $r_K^{2K} \le e^{-2K\eta_K} \to 0$ and $1 - r_K^2 = \eta_K(2-\eta_K) \sim 2K^{-1/2}$. Each term in the average is therefore asymptotic to $1/(2\sqrt{K})$:

$$\boxed{\overline{G}_K^{\mathrm{SGD}} = \frac{1+o(1)}{\sqrt{K}}.} \tag{D.3}$$

The optimized BV bound is $(1+\sqrt{1+4K})/(2K)$, so both its rate and leading constant are attained by this prescribed run.

For TSGD, use $\eta = 1$ and any square-integrable $g_0 = 1 + e_0$. Exact differences give $D_{k+1} = x_{k+1} - x_k = -g_k$; therefore $g_1 = 0$ and $x_k = -e_0$ for all $k \ge 1$. Hence

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} = \frac{1}{K} + \left(1 - \frac{1}{K}\right) E_0.} \tag{D.4}$$

With exact initialization, TSGD reaches the minimizer after one step. For fixed $E_0 > 0$, horizon-tuned SGD eventually has the smaller error. The comparison depends on initialization as well as transport.

D.2 Multiplicative Noise: Fresh Gradients and False Convergence

For (3.3), $\mathbb{E}[a] = 1$ and $\mathbb{E}[(a-1)^2] = 1$. The oracle is unbiased and its pointwise noise variance is x^2, so global BV does not hold. Coupled centered differences nevertheless satisfy

$$\mathbb{E}[|(a-1)(x-y)|^2] = |x-y|^2,$$

giving $\delta = 1$. These comparisons use exact dynamics rather than the BV theorem.

At $\eta = 1/2$, SGD obeys $x_{k+1} = (1 - a_k/2)x_k$; it remains at one until the first sample $a = 2$, then stays at zero. Thus

$$\boxed{G_k^{\mathrm{SGD}} = 2^{-k}, \qquad \overline{G}_K^{\mathrm{SGD}} = \frac{2(1-2^{-K})}{K}.} \tag{D.5}$$

For TSGD at the same step, let $g_0 = \nabla f(x_0) = 1$ and $N := \min\{n \ge 1 : a_n = 2\}$. The estimator recursion is $g_{k+1} = (1 - a_{k+1}/2)g_k$, so

$$\mathbb{P}(N = n) = 2^{-n}, \qquad g_k = \mathbf{1}_{\{k<N\}}, \qquad x_k = 1 - \frac{1}{2}\min\{k, N\}.$$

Since $N < \infty$ almost surely and $\mathbb{E}[N^2] = 6$, dominated convergence gives $x_k \to 1 - N/2$ in mean square. Using $\mathbb{E}[N] = 2$,

$$\boxed{\lim_{k\to\infty} G_k^{\mathrm{TSGD}} = \mathbb{E}[(1 - N/2)^2] = \frac{1}{2}.} \tag{D.6}$$

The estimator becomes zero almost surely in finite time, but the limiting true gradient is nonzero with probability $1 - \mathbb{P}(N = 2) = 3/4$. Cesàro averaging also gives $\overline{G}_K^{\mathrm{TSGD}} \to 1/2$.

D.3 Sharpness for Horizon-Tuned TSGD

For the same oracle with exact initialization, (A.4) gives $\eta_K^\star = 2/(1 + \sqrt{4K - 3})$, since $L = \delta = 1$ and $\Delta_0 = 1/2$. The optimized guarantee becomes

$$\overline{G}_K^{\mathrm{TSGD}} \le \frac{1 + \sqrt{4K - 3}}{2K}. \tag{D.7}$$

Lemma D.1 (Asymptotic sharpness for tuned TSGD)**.** *For (3.3), with exact initialization and the constant step $\eta_K^\star$ for the K-iteration run,*

$$\lim_{K\to\infty} \sqrt{K}\, \overline{G}_K^{\mathrm{TSGD}} = 1.$$

Proof. Fix $K \ge 2$ and write $\eta = \eta_K^\star$. The updates imply

$$g_{k+1} = (1 - \eta a_{k+1}) g_k, \qquad e_{k+1} = e_k + \eta(1 - a_{k+1}) g_k.$$

Set $C_k := \mathbb{E}[g_k e_k]$ and $q := \mathbb{E}[(1 - \eta a)^2] = 1 - 2\eta + 2\eta^2$. Independence yields

$$M_{k+1} = q M_k, \qquad E_{k+1} = E_k + \eta^2 M_k, \qquad C_{k+1} = (1 - \eta) C_k + \eta^2 M_k.$$

Starting from $M_0 = 1$ and $E_0 = C_0 = 0$, these give

$$M_k = q^k, \qquad E_k = \frac{\eta}{2(1 - \eta)}(1 - q^k), \qquad C_k = \eta^2 \sum_{t=0}^{k-1} (1 - \eta)^{k-1-t} q^t.$$

Since $0 < \eta < 1$ and $q < 1$, all summands are nonnegative and

$$\sum_{k<K} C_k \le \eta^2 \left(\sum_{t\ge 0} q^t \right) \left(\sum_{j\ge 0} (1 - \eta)^j \right) = \frac{\eta^2}{(1 - q)\eta} = \frac{1}{2(1 - \eta)}.$$

Thus the average cross moment is $O(K^{-1})$ as $\eta_K^\star \to 0$. For the other moments,

$$\frac{1}{K} \sum_{k<K} M_k = \frac{1 - q^K}{K(1 - q)}, \qquad \frac{1}{K} \sum_{k<K} E_k = \frac{\eta}{2(1 - \eta)} \left(1 - \frac{1}{K} \sum_{k<K} M_k \right).$$

Now $\eta_K^\star \sim K^{-1/2}$, $1 - q = 2\eta_K^\star(1 - \eta_K^\star) \sim 2K^{-1/2}$, and $q^K \le e^{-K(1-q)} \to 0$. Therefore

$$\frac{1}{K} \sum_{k<K} M_k \sim \frac{1}{2\sqrt{K}}, \qquad \frac{1}{K} \sum_{k<K} E_k \sim \frac{1}{2\sqrt{K}}.$$

Finally, $x_k = g_k - e_k$ implies $G_k = M_k + E_k - 2C_k$. The averaged cross term is lower order, which proves the claim. □

The theorem's bound has the correct leading rate and constant for this prescribed TSGD run. Fixed-step SGD in (D.5) has average error $\Theta(K^{-1})$, whereas this horizon-tuned TSGD run has $\Theta(K^{-1/2})$ error. This is not a lower bound over all TSGD stepsize schedules.

D.4 Fixed Variance and Variable Similarity

Lemma D.2 (Fixed variance and variable similarity)**.** *For (3.1), the noise is centered, satisfies (3.2), and has smallest global similarity constant $\delta = \sigma\omega$ for $\omega \geq 0$.*

Proof. A uniform phase makes the cosine have mean zero and second moment $1/2$, so $\mathbb{E}[r(x,\varphi)^2] = 2\sigma^2(1/2) = \sigma^2$. For the coupled difference,

$$r(x,\varphi) - r(y,\varphi) = -2\sqrt{2}\sigma \sin\left(\frac{\omega(x+y)}{2} + \varphi\right) \sin\left(\frac{\omega(x-y)}{2}\right).$$

Squaring and using $\mathbb{E}_\varphi[\sin^2(c+\varphi)] = 1/2$ for every fixed c gives

$$\mathbb{E}[|r(x,\varphi) - r(y,\varphi)|^2] = 8\sigma^2 \cdot \frac{1}{2} \sin^2\left(\frac{\omega(x-y)}{2}\right) = 4\sigma^2 \sin^2\left(\frac{\omega(x-y)}{2}\right).$$

This verifies (3.2). The inequality $|\sin t| \leq |t|$ proves similarity with $\delta = \sigma\omega$; minimality follows from

$$\lim_{y\to x} \frac{4\sigma^2 \sin^2(\omega(x-y)/2)}{(x-y)^2} = \sigma^2\omega^2.$$

□

SGD's fresh noise has conditional variance σ^2 at every iterate, giving the frequency-independent exact recursion

$$G_{k+1}^{\mathrm{SGD}} = (1 - L\eta)^2 G_k^{\mathrm{SGD}} + L^2\eta^2\sigma^2. \tag{D.8}$$

Indeed, even its conditional noise distribution is independent of x_k and ω, since adding ωx_k to a uniform phase preserves its distribution modulo 2π. For TSGD,

$$\begin{aligned} E_{k+1} - E_k &= 4\sigma^2 \mathbb{E}\left[\sin^2\left(\frac{\omega(x_{k+1} - x_k)}{2}\right)\right] \\ &\leq \sigma^2\omega^2\eta^2 M_k. \end{aligned} \tag{D.9}$$

For fixed $\sigma > 0$, varying ω realizes any smallest global similarity constant without changing the pointwise noise distribution. The sine factor also shows that increasing ω need not increase every realized increment's error monotonically.

D.5 Numerical Illustration of Fixed Variance and Variable Similarity

Numerical protocol and reproducibility. Figure 4 uses $L = \sigma = x_0 = 1$, $\eta = 0.1$, and $\omega \in \{0, 0.5, 1, 2\}$. Each curve averages 100,000 independent trajectories, generated with NumPy's default random generator from seed 20261023 and separate spawned random streams for each curve. SGD is simulated at $\omega = 1$; its law is frequency-independent. All later phases are independent uniform samples; only the two endpoint evaluations within a transport increment share their phase.

Initialization and cost. This controlled oracle allows an antithetic initial pair:

$$g_0 = \frac{1}{2}(h(x_0,\varphi_0) + h(x_0,\varphi_0+\pi)) = Lx_0.$$

We charge two pointwise evaluations for this exact estimate for every method. This uses a specific sampling symmetry, not a claim that two independent samples generally yield exact initialization.

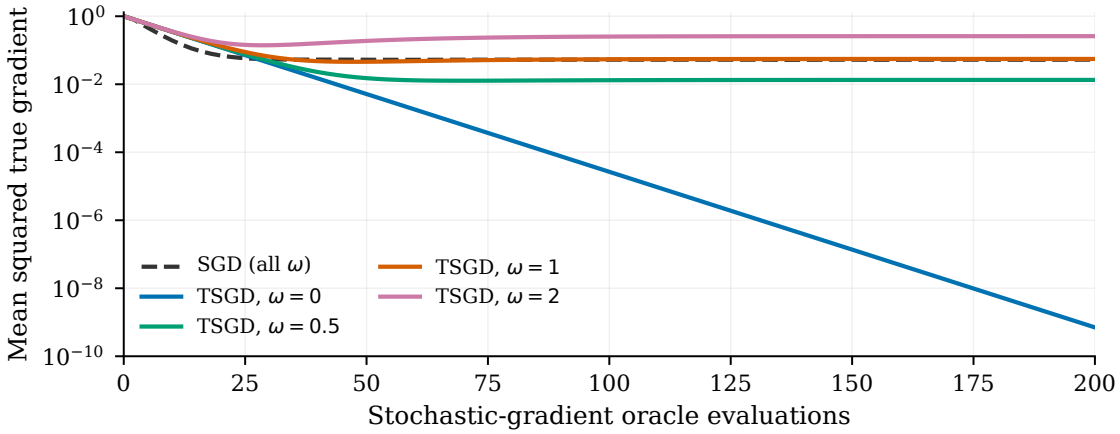


Figure 4: Same pointwise variance, different transport accuracy. Monte Carlo means over 10^5 runs, with bands of two standard errors; $L = \sigma = x_0 = 1$ and a common constant stepsize $\eta = 0.1$. Every method initializes by averaging gradients at phases φ_0 and $\varphi_0 + \pi$: their noises cancel, giving $g_0 = Lx_0$. These two evaluations are included in the horizontal-axis count. Subsequent SGD updates cost one evaluation and TSGD updates cost two. SGD is simulated at $\omega = 1$; its expected curve is identical at all frequencies. The sampling, cost accounting, and numerical checks are described below.

All methods use this estimate for their first update. Thereafter, SGD uses one new stochastic gradient per update and TSGD uses two pointwise evaluations per coupled difference. After $k \geq 1$ iterate updates, the costs are $k+1$ for SGD and $2k$ for TSGD. No unused final transport correction is computed. Hence a budget of 200 permits 199 SGD updates and 100 TSGD updates. This protocol isolates transport accuracy after an equally charged, identical initial estimate; it does not compare generic independent-batch initialization procedures.

Stepsize and analytical checks. The common stepsize is held fixed for all frequencies and methods. For the sinusoidal quadratic, conditional centeredness and similarity give

$$\mathbb{E}[g_{k+1}^2 \mid \mathcal{F}_k] \leq \big((1-\eta)^2 + \eta^2\omega^2\big) g_k^2 \leq 0.85 g_k^2.$$

Thus the chosen step is mean-square stable across all tested frequencies. It illustrates constant-step error accumulation, not the horizon-optimized guarantee of Section 4. The SGD recurrence (D.8) applies after the common first update, for which $G_1 = 0.9^2$. Every simulated SGD mean is within 3.35 estimated standard errors of this exact recurrence. At $\omega = 0$, TSGD follows $G_k = 0.9^{2k}$; the relative numerical discrepancy is below 10^{-12} throughout the plotted run.

Means and tail contributions. At budget 200, the final sample means and standard errors are

Method	Mean squared true gradient	Standard error
SGD	0.052791	0.000226
TSGD, $\omega = 0$	7.0551×10^{-10}	below 10^{-20}
TSGD, $\omega = 0.5$	0.013409	0.000063
TSGD, $\omega = 1$	0.055485	0.000331
TSGD, $\omega = 2$	0.259610	0.003050

All trajectories are retained. For $\omega = 2$, the largest 0.1% of final squared gradients contribute 7.72% of their sum; the largest observed value is 77.18. The reported means therefore include the comparatively rare large errors rather than summarizing only typical trajectories. The shaded bands are the sample mean plus or minus two estimated standard errors at each cost, not simultaneous confidence bands or trajectory ranges.

The script `scripts/similarity_experiment.py` reproduces the simulations, analytical checks, and figure. Its dependencies are recorded in `scripts/requirements.txt`. The accompanying `figures/similarity_experiment.csv` records per-iterate means and standard errors; the `.json` file records parameters, software versions, analytical checks, and final quantile summaries.

D.6 Least-Squares Experiments

Model and protocol. All least-squares plots use $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$ in $d = 10$ dimensions. Let $H = \mathrm{diag}(0.1, 0.2, \ldots, 1)$, $v = 1 + \kappa\zeta$ with $\mathbb{P}(\zeta = 1) = \mathbb{P}(\zeta = -1) = 1/2$, $A = \sqrt{v}\, H^{1/2}$, and $b = Ax^\star + s\varepsilon$, where $x^\star = 0$ and $\varepsilon \sim \mathcal{N}(0, I_d)$ is independent of v. Thus $\nabla f(x) = H(x - x^\star)$, the stochastic gradient is $h(x; v, \varepsilon) = vH(x - x^\star) - s\sqrt{v}\, H^{1/2}\varepsilon$, and the global centered-similarity constant is $\delta = \kappa$ since $L = \|H\| = 1$. The initial point is $x_0 = \mathbf{1}/\|H\mathbf{1}\|$, so $\|\nabla f(x_0)\|^2 = 1$. All panels use $\eta = 0.5$, 4,000 independent trajectories, 400 updates, and seed 20261028; plotted bands are pointwise means ± 2 Monte Carlo standard errors.

Panels A and B use sampled initial estimates. Panel A has $s = 1$ and $\kappa = 0.05$; TSGD averages $B_0 \in \{1, 128, 512\}$ independent stochastic gradients for initialization. Its initial mean-square error is

$$E_0 = \frac{\kappa^2 \|Hx_0\|^2 + s^2 \,\mathrm{tr}(H)}{B_0},$$

so increasing B_0 lowers the initialization contribution, while a transport-error residual remains. The plotted floor falls from about 5.51 at $B_0 = 1$ to 0.044 and 0.011 at $B_0 = 128$ and 512. Panel B has $s = 0$ and $\kappa = 0.9$, with TSGD initialized from $B_0 = 128$ samples. Its estimator and SGD's true-gradient norm have the same coordinatewise geometric contraction factors; their decay curves nearly overlap because the batch initialization adds only 0.63% to the initial second moment. The TSGD true-gradient norm nevertheless approaches about 0.268, giving the false-convergence illustration in Figure 2(b).

Panels C and D instead use controlled initializations and are not computational-cost comparisons. In C, both methods start from the exact gradient and use matched feature draws across $s \in \{0, 1, 5\}$, with $\kappa = 0.2$. The shared label-noise term cancels from each TSGD increment, so its trajectories coincide across all three values of s. In D, TSGD starts from $g_0 = Hx_0 + e_0$, where independently sampled $e_0 \sim \mathcal{N}(0, (E_0/d)I_d)$; the four choices are $\kappa \in \{0.05, 0.9\}$ and $E_0 \in \{10^{-3}, 10^{-1}\}$. The script `experiments/least_squares.py` reproduces the runs and figures.

Initialization quality. Figure 5 displays panel A. The residual decreases as the initial batch becomes more accurate, while transport error creates an additional floor even with a highly accurate initialization. A single sample gives a worse residual than SGD here; larger initialization batches give substantially smaller residuals. This illustrates that transport preserves the quality of its starting estimate rather than correcting it.

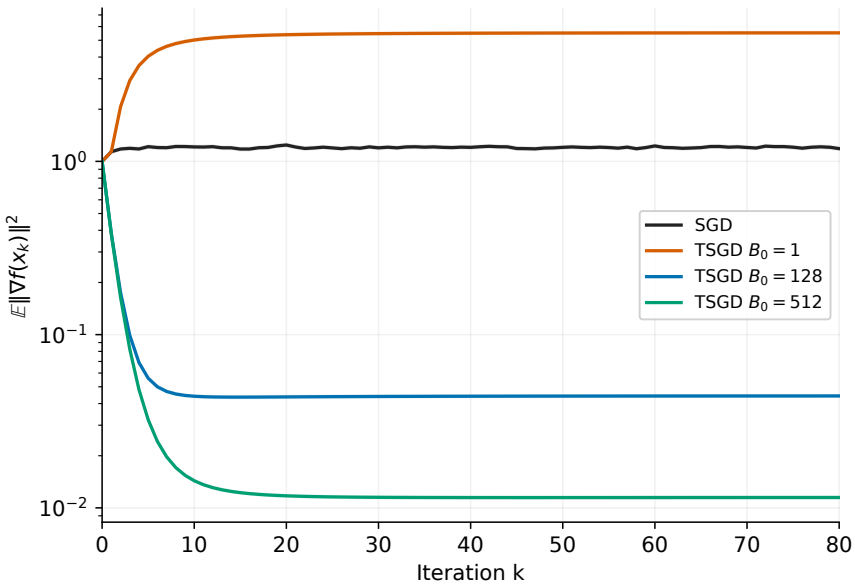


Figure 5: Initialization quality on vector least squares with noisy labels. TSGD is initialized by averaging B_0 independent samples. Means over 4,000 trajectories with pointwise bands of ± 2 standard errors.

E Oracle Complexity and Additional Algorithmic Details

E.1 Initialization and Accuracy Targets

Suppose an unbiased oracle at x_0 has variance at most σ_0^2. With B_0 independent samples, independent also of subsequent transport samples, set

$$g_0 = \frac{1}{B_0}\sum_{i=1}^{B_0} h(x_0, \xi_i^{(0)}).$$

Independence and centering cancel the cross terms, giving

$$E_0 = \mathbb{E}[\|g_0 - \nabla f(x_0)\|^2] \le \frac{\sigma_0^2}{B_0}.$$

Thus $B_0 \ge \max\{1, \lceil 3\sigma_0^2/\varepsilon^2 \rceil\}$ ensures $E_0 \le \varepsilon^2/3$, as required by Corollary A.1. Only the variance at x_0 is needed for this construction. The main theorem itself requires neither unbiased initialization nor a pointwise oracle.

Count K iterate updates. Single-sample SGD uses K pointwise gradient evaluations, whereas the shared-sample TSGD construction uses at most $B_0 + 2(K-1)$. Its final correction D_K is unused: g_{K-1} suffices to compute x_K, and the output is selected from $x_0, \dots, x_{K-1}$. Under the common BV assumption, one can take $\sigma_0^2 \le \sigma^2$. Combining the batch size above with (A.10) and (A.11) gives sufficient evaluation budgets of order

$$\begin{aligned} N_{\mathrm{SGD}}(\varepsilon) &= O\left(1 + \frac{L\Delta_0}{\varepsilon^2} + \frac{L\Delta_0\sigma^2}{\varepsilon^4}\right), \\ N_{\mathrm{TSGD}}(\varepsilon) &= O\left(1 + \frac{\sigma^2}{\varepsilon^2} + \frac{L\Delta_0}{\varepsilon^2} + \frac{\delta^2\Delta_0^2}{\varepsilon^4}\right). \end{aligned} \tag{E.1}$$

These are upper bounds for the stated implementations, not oracle lower bounds. The initialization term depends on this choice of g_0 and can change when other initial gradient information is available.

E.2 Comparing Bounds at Equal Iteration Counts

Suppose both BV and global δ-similarity hold for the same oracle, both methods start at the same deterministic x_0, and $\Delta_0 > 0$. Define the exact optimized bounds

$$\begin{aligned} U_{\mathrm{SGD}}(K) &:= \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}}\right), \\ U_{\mathrm{TSGD}}(K) &:= \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + 4(K-1)\delta^2}\right) + E_0. \end{aligned}$$

Subtracting shows that $U_{\mathrm{TSGD}}(K) < U_{\mathrm{SGD}}(K)$ holds exactly when

$$E_0 < \frac{\Delta_0}{K}\left[\sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}} - \sqrt{L^2 + 4(K-1)\delta^2}\right]. \tag{E.2}$$

With exact initialization this reduces to

$$2(K-1)\delta^2\Delta_0 < KL\sigma^2. \tag{E.3}$$

For fixed positive δ, σ, Δ_0 and $E_0 = o(K^{-1/2})$, the leading terms favor the TSGD bound when $2\delta^2\Delta_0 < L\sigma^2$. In contrast, a fixed positive E_0 remains in the TSGD bound while the tuned SGD bound tends to zero. These inequalities compare upper bounds; they do not imply the same ordering of actual stationarity errors.

The powers of the stepsize in intermediate calculations should not be compared directly. For SGD, the one-step descent penalty is $L\eta^2\sigma^2/2$; summing and dividing by the descent coefficient gives $L\eta\sigma^2/(2 - L\eta)$ in (A.6). For TSGD, $\delta^2\eta^2 M_k$ instead bounds the increase $E_{k+1} - E_k$ in tracking error. This increment persists in subsequent estimates and receives the weight $K - 1 - k$ in (A.1). The optimized bounds account for this accumulation.

There is also a fixed-step comparison when $\delta = 0$: TSGD may use $\eta = 1/L$ for all horizons, obtaining $2L\Delta_0/K + E_0$, while SGD's corresponding bound is $2L\Delta_0/K + \sigma^2$. For $\delta > 0$, the sufficient condition of Theorem 4.1 makes the prescribed TSGD step horizon dependent. This does not imply that every fixed-step TSGD run must fail.

E.3 Comparing Bounds at Equal Oracle Budgets

Let B_0 denote the cost of a chosen initialization with error E_0, without imposing a batching or unbiasedness requirement. With the cost convention above, an integer budget $N \geq \max\{1, B_0\}$ allows

$$K_{\mathrm{SGD}} = N, \qquad K_{\mathrm{TSGD}} = 1 + \left\lfloor \frac{N - B_0}{2} \right\rfloor. \tag{E.4}$$

The bounds to compare are therefore $U_{\mathrm{SGD}}(N)$ and $U_{\mathrm{TSGD}}(K_{\mathrm{TSGD}})$, including the error of the chosen initialization in the latter.

For fixed positive δ, σ, Δ_0, suppose $B_0 = o(N)$ and $E_0 = o(N^{-1/2})$. Then $K_{\mathrm{TSGD}} \sim N/2$ and

$$U_{\mathrm{SGD}}(N) \sim \frac{\sqrt{2L\Delta_0\sigma^2}}{\sqrt{N}}, \qquad U_{\mathrm{TSGD}}(K_{\mathrm{TSGD}}) \sim \frac{2\sqrt{2}\delta\Delta_0}{\sqrt{N}}.$$

The leading TSGD bound is smaller when $4\delta^2\Delta_0 < L\sigma^2$, compared with the equal-iteration threshold $2\delta^2\Delta_0 < L\sigma^2$. A fixed positive initialization error does not satisfy these asymptotic conditions. Exact initialization must also have its cost counted in B_0. Direct increment oracles and implementations that evaluate endpoints jointly require their own cost accounting.

E.4 Refreshing Improves Complexity on a Common Problem Class

Common assumptions and accuracy convention. Fix positive bounds $\Delta, L, \sigma, \delta$, and consider lower-bounded objectives with initial gap at most Δ and L-Lipschitz gradients, together with unbiased stochastic gradient oracles satisfying both BV and centered δ-similarity. These assumptions control individual gradients and their differences simultaneously. Indeed, unbiasedness gives

$$\begin{aligned} \mathbb{E}[\|h(x,\xi) - h(y,\xi)\|^2] &= \|\nabla f(x) - \nabla f(y)\|^2 + \mathbb{E}[\|r(x,\xi) - r(y,\xi)\|^2] \\ &\leq (L^2 + \delta^2)\|x - y\|^2. \end{aligned} \tag{E.5}$$

Thus the oracle is mean-squared smooth with constant $\bar{L} = \sqrt{L^2 + \delta^2}$. We compare the number of stochastic-gradient evaluations needed to find $\mathbb{E}[\|\nabla f(\widehat{x})\|] \leq \varepsilon$, the criterion used by Arjevani et al. [2], SPIDER, and PAGE. Corollary A.1 uses the stronger criterion $\mathbb{E}[\|\nabla f(\widehat{x})\|^2] \leq \varepsilon^2$, which implies it by Jensen's inequality. The accuracy parameter is therefore on the same gradient-norm scale, and the lower bound below applies to both criteria. The oracle model allows gradients to be evaluated at multiple points using the same sample, which is precisely the access TSGD uses to form its differences.

Applicable upper bounds. For fixed positive parameter bounds, (E.1) gives $O(\varepsilon^{-4})$ oracle budgets for SGD and pure TSGD, including initialization. Mean-squared smoothness and BV also ensure that the online guarantees of SPIDER [6] and PAGE [15] apply, yielding $O(\varepsilon^{-3})$ complexity with their prescribed batch sizes and refresh schedules.

A lower bound on the same class. The upper bounds show that $O(\varepsilon^{-3})$ evaluations suffice. To justify optimality of this exponent, the lower bound must apply with the same separately prescribed bounds L and δ. Theorem 3.2 of Fradin et al. [7] provides such a bound for zero-respecting algorithms. Starting at the origin, these algorithms only introduce coordinates revealed by previous oracle responses. This class includes the standard gradient-based implementations discussed here, but does not include every randomized algorithm.

Corollary E.1 (Specialization of Fradin et al., Theorem 3.2)**.** *Fix $\Delta, L, \sigma, \delta > 0$. For all sufficiently small $\varepsilon > 0$, any zero-respecting algorithm that guarantees $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \varepsilon$ uniformly over the class satisfying Assumptions A1–A4 and $f(x_0) - f_{\inf} \le \Delta$ requires*

$$\Omega\left(\frac{\Delta\delta\sigma}{\varepsilon^3}\right) \tag{E.6}$$

stochastic-gradient evaluations in the worst case. The oracle permits any fixed finite number of simultaneous query points sharing the same sample, with independent samples across rounds. The same lower bound holds for the stronger target $\mathbb{E}[\|\nabla f(\widehat{x})\|^2] \le \varepsilon^2$.

Proof. Set $p = q = 2$ in Theorem 3.2 of Fradin et al. [7]. Their bounded central moment and (q, δ)-similarity assumptions then become BV and centered δ-similarity, respectively. With $L_1 = L$ and $\sigma_1 = \sigma$, their lower bound is

$$\Omega\left(\min\left\{\frac{L\Delta}{\varepsilon^2} + \frac{L\Delta\sigma^2}{\varepsilon^4}, \quad \frac{\sigma^2 + (L+\delta)\Delta}{\varepsilon^2} + \frac{\delta\Delta\sigma}{\varepsilon^3}\right\}\right).$$

The theorem applies for $0 < \varepsilon \le c_1\sqrt{L\Delta}$ with a universal $c_1 > 0$. For fixed positive parameters, the first branch grows as $L\Delta\sigma^2/\varepsilon^4$, whereas the second is asymptotic to $\delta\Delta\sigma/\varepsilon^3$. Their ratio tends to infinity, so the minimum is the second branch for sufficiently small ε, giving the claimed bound. The lower bound counts shared-sample query rounds, each requiring at least one gradient evaluation, so it also bounds the evaluation count. Finally, $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \sqrt{\mathbb{E}[\|\nabla f(\widehat{x})\|^2]}$ extends the bound to the squared-norm criterion. □

Together with the SPIDER/PAGE upper bounds, Corollary E.1 establishes optimality of the ε^{-3} exponent among zero-respecting algorithms on this common class as $\varepsilon \to 0$, with the positive bounds $\Delta, L, \sigma, \delta$ held fixed. This does not establish an $\Omega(\varepsilon^{-4})$ lower bound for pure TSGD; its $O(\varepsilon^{-4})$ guarantee need not be tight, and exact transport can already yield faster convergence.

E.5 Additional Estimator Identities and Sampling Details

We expand the relationships in Table 3. Throughout, H_{k+1} is a fresh unbiased estimate of $\nabla f(x_{k+1})$, possibly a minibatch average or an exact gradient, and D_{k+1} estimates $\nabla f(x_{k+1}) - \nabla f(x_k)$. Unless stated otherwise, the two endpoints of a stochastic difference use the same fresh sample or minibatch. These are identities for the gradient estimators; each method has its own stepsizes, output rules, and iterate update.

E.5.1 Periodic Restarts: SARAH and SPIDER

SARAH initializes an inner loop with a full gradient and then applies $g_{k+1} = g_k + D_{k+1}$ using stochastic gradient differences [16, 17]. SARAH-IN is therefore a transport phase with exact initialization. Inexact SARAH replaces the full gradient by a sufficiently accurate minibatch estimate [18]. SPIDER uses the same recursive estimator between periodic full-gradient or large-batch recomputations [6]; normalized or adaptive iterate updates are additional features of that method.

For a transport phase starting at iteration s, conditional unbiasedness gives

$$E_k = E_s + \sum_{t=s}^{k-1} \mathbb{E}[\|d_{t+1}\|^2]. \tag{E.7}$$

Indeed, $e_{t+1} = e_t + d_{t+1}$ and $\mathbb{E}[\langle e_t, d_{t+1}\rangle] = 0$ give $E_{t+1} = E_t + \mathbb{E}[\|d_{t+1}\|^2]$, which telescopes over the phase. At a restart, replacing the estimator by an exact gradient sets $E_s = 0$; a fresh independent batch of size B gives $E_s \le \sigma^2/B$ under BV. Tracking errors then accumulate only within the new phase. Shorter phases limit this accumulation but require more frequent recomputations. A restart corrects the estimate at the current point; it does not undo the effect of earlier errors on the iterates.

E.5.2 Random Restarts: Geom-SARAH

Geom-SARAH retains the SARAH recursion but draws the inner-loop length from a geometric distribution [10]. Each epoch starts with a fresh batch estimate and passes its final point to the next epoch. Batch sizes and expected epoch lengths may vary between epochs. If N_j is the number of inner steps in epoch j, write, for $0 < p_j \le 1$,

$$\mathbb{P}(N_j = n) = p_j(1-p_j)^n, \quad n \ge 0, \qquad \mathbb{E}[N_j] = \frac{1-p_j}{p_j}.$$

The memoryless property corresponds to a constant stopping probability within an epoch. With initialization batch B_j and transport batch b_j, the expected number of pointwise gradient evaluations is at most $B_j + 2b_j\mathbb{E}[N_j]$. Longer phases spread the initialization cost over more steps but allow errors to accumulate for longer. The geometric law controls the expected duration without imposing a deterministic maximum. Geom-SARAH couples this randomization with batch-size schedules in its convergence analysis.

E.5.3 Probabilistic Refreshing: PAGE and MARINA

For a fixed refresh probability p, PAGE's estimator is [15]

$$g_{k+1} = c_{k+1}H_{k+1} + (1-c_{k+1})(g_k + D_{k+1}), \qquad c_{k+1} \sim \text{Bernoulli}(p). \tag{E.8}$$

The decision is independent of the history and the new samples. The refresh branch uses a fresh batch, and the transport branch typically uses a smaller batch of coupled differences. Define

$$q_k = \mathbb{E}[\|d_{k+1}\|^2], \qquad v_{k+1} = \mathbb{E}[\|H_{k+1} - \nabla f(x_{k+1})\|^2].$$

Conditioning on the branch and using conditional unbiasedness of the transport increment yields

$$E_{k+1} = (1-p)(E_k + q_k) + p\, v_{k+1}. \tag{E.9}$$

There is no cross term between the two branches because exactly one is selected. Within the transport branch, the cross term between e_k and d_{k+1} vanishes by conditional unbiasedness. Each transport error survives only until the next refresh. An exact refresh has $v_{k+1} = 0$; a stochastic refresh replaces the previous error by its own batch error. The boundary $p = 0$ recovers TSGD, while $p = 1$ gives minibatch SGD. For $0 < p < 1$, refreshing mitigates persistent initialization error and accumulated transport errors.

MARINA uses the same probabilistic choice in distributed optimization [9]. For $f = n^{-1} \sum_i f_i$, the refresh branch computes $H_{k+1} = \nabla f(x_{k+1})$ and the transport branch uses compressed local gradient differences:

$$D^{\mathcal{Q}}_{k+1} = \frac{1}{n} \sum_{i=1}^{n} \mathcal{Q}_{i,k+1}(\nabla f_i(x_{k+1}) - \nabla f_i(x_k)) \,. \tag{E.10}$$

Unbiased compression makes this increment conditionally unbiased, although the full gradient estimator need not be conditionally unbiased. Refreshing removes accumulated compression error; transporting communicates compressed changes. Thus increment noise can arise from compression as well as stochastic gradient sampling. Verifying a similarity bound for the compressed increment requires the corresponding compression and local-smoothness assumptions; unbiased compression alone only establishes the conditional mean.

E.5.4 Mixing Fresh and Transported Estimates: MVR and STORM

The momentum variance-reduction recursion used by STORM combines a fresh estimate with a transported estimate at every iteration [4]. For fresh weight $\alpha \in [0, 1]$,

$$\begin{aligned} g_{k+1} &= (1 - \alpha)(g_k + D_{k+1}) + \alpha H_{k+1} \\ &= h(x_{k+1}, \xi_{k+1}) + (1 - \alpha)(g_k - h(x_k, \xi_{k+1})). \end{aligned} \tag{E.11}$$

The second equality uses the same fresh sample ξ_{k+1} in H_{k+1} and D_{k+1}. STORM chooses an iteration-dependent weight linked to its stepsize; the fixed-weight notation exposes the estimator identity. Subtracting the true gradient gives

$$e_{k+1} = (1 - \alpha)e_k + (1 - \alpha)d_{k+1} + \alpha(H_{k+1} - \nabla f(x_{k+1})). \tag{E.12}$$

Mixing damps inherited error while introducing transport and fresh-gradient noise. A larger α strengthens the correction but places more weight on pointwise noise; a smaller α relies more on accurate changes. The boundaries $\alpha = 0$ and $\alpha = 1$ give TSGD and SGD, respectively.

For a deterministic weight, conditional unbiasedness eliminates cross terms with e_k, so the precise mean-square relation is

$$\begin{aligned} E_{k+1} = (1 - \alpha)^2(E_k + q_k) + \alpha^2 v_{k+1} \\ + 2\alpha(1 - \alpha)\mathbb{E}[\langle d_{k+1}, H_{k+1} - \nabla f(x_{k+1})\rangle] \,. \end{aligned}$$

Unlike PAGE's mutually exclusive branches, the two new noises occur together and share a sample. Their covariance generally cannot be discarded. If weights depend on the history, the analogous equality holds conditionally, with the weights retained inside expectations after averaging over that history.

E.5.5 Transport Corrections within Momentum: MARS and VENUS

Before clipping, MARS scales the gradient-difference correction within momentum [23]. Write $\beta = 1 - \beta_1$ for the fresh-gradient weight, where β_1 is its original momentum coefficient. The estimator is

$$g_{k+1} = (1-\beta)(g_k + \gamma D_{k+1}) + \beta H_{k+1}. \tag{E.13}$$

The difference uses the same sample at both endpoints. Setting $\gamma = 0$ yields MSGD's exponential moving average; $\gamma = 1$ yields MVR with fresh weight β. For $\gamma \in [0, 1]$, the VENUS manuscript gives the convex interpolation [1]

$$g_{k+1}^{\mathrm{MARS}} = (1-\gamma) g_{k+1}^{\mathrm{MSGD}} + \gamma g_{k+1}^{\mathrm{MVR}},$$

where the two candidates share the incoming estimate, sample, and fresh weight β. The expanded identity gives pure transport at $(\beta, \gamma) = (0, 1)$, although MARS's original rescaled correction is undefined at $\beta = 0$.

For $\gamma \neq 1$, the conditional mean of γD_{k+1} is $\gamma(\nabla f(x_{k+1}) - \nabla f(x_k))$, so Assumption A5 generally fails. Clipping and preconditioning further change the algorithm. A difference using a cached gradient can violate both conditional unbiasedness and similarity (Assumptions A5–A6).

VENUS further separates the samples used by the MSGD and MVR branches [1]. It preserves shared sampling within each transported difference but assigns part of the fresh contribution to an independent batch. This changes the covariance between refreshing and transporting as well as their weights. The estimator design therefore depends on sample coupling, not only on the coefficients in a linear combination.

E.5.6 Pure Transport as an Established Special Case

The transport mechanism has also been studied independently of refreshing, including SARAH-IN and iSARAH-IN [17, 18] and PAGE-NS [21]. The $p = 0$ boundary of PAGE is the same recursion [15]. The recursive-gradient (RG) estimator used within the inexact composite gradient method (I-CGM) also reduces to pure transport at its $\beta = 0$ boundary [11]. Here β is the RG mixing parameter, distinct from MARS's fresh weight. The shared recursion does not imply identical iterate updates or convergence assumptions.